Estimating a Firm's Cost of Capital

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Chapter Overview

In earlier chapters, we referred to the *discount rate* simply as the interest rate used to calculate present values. To describe the appropriate discount rate that should be used to calculate the value of an investment's future cash flows, financial economists use terms like *opportunity cost of capital* or simply *cost of capital.* In this chapter, we focus our attention on the firm's overall weighted average cost of capital (or WACC), which is the discount rate that should be used to value an entire firm. In addition to describing the WACC, we consider the determinants of its various components—the cost of debt and the cost of equity.

1 Introduction

The firm's **weighted average cost of capital** (or **WACC,** pronounced "whack") is the weighted average of the expected after-tax rates of return of the firm's various sources of capital. As we will discuss in this chapter, the WACC is the discount rate that should be used to discount the firm's expected free cash flows to estimate firm value.

A firm's WACC can be viewed as its **opportunity cost of capital,** which is the expected rate of return that its investors forgo from alternative investment opportunities with equivalent risk. This may sound like "finance-speak," but it illustrates a very important concept. Because investors could invest their money elsewhere, providing money to a firm by purchasing its securities (bonds and shares of stock) has an opportunity cost. That is, if an investor puts her money into the stock of Google (GOOG), she gives up (forgoes) the return she could have earned by investing in Microsoft (MSFT) stock. What this means is that if Google and Microsoft have equivalent risks, the expected rate of return on Microsoft stock can be viewed as the opportunity cost of capital for Google, and the expected return on Google stock can be viewed as the opportunity cost of capital for Microsoft.

In addition to providing the appropriate discount rate to calculate the firm's value, firms regularly track their WACC and use it as a benchmark for determining the appropriate discount rate for new investment projects, for valuing acquisition candidates, and for evaluating their own performance.

2 Value, Cash Flows, and Discount Rates

Figure 1 reviews the three-step DCF valuation methodology. Up until this point, we have focused entirely on project valuation (the right-hand column). We now introduce equity valuation (the middle column) and firm valuation.

The key learning point from the second and third columns is that cash flow calculations and discount rates must be properly aligned. If you are trying to estimate the value of the equity invested in the project (i.e., **equity value**), then you will estimate equity free cash flows and use a discount rate that is appropriate for the equity investors. On the other hand, if you are estimating the value of an entire firm, which equals the value of the combined equity and debt claims, then the appropriate cash flow is the combination of debt and equity cash flows. In that case, the appropriate discount rate is a combination of the debt- and equity-holder rates, or what we will refer to as the weighted average cost of capital.

It should be emphasized that in all of these cases, we are assuming that the estimated cash flows are what we have defined as expected cash flows. If we are using conservative cash flows, then we would want to use a lower discount rate. Similarly, if the estimated cash flows are really "hoped-for" or optimistic cash flows, then we will need a higher discount rate to offset the optimistic cash flow forecasts.

Defining a Firm's WACC

The weighted average cost of capital (WACC) is a weighted average of the after-tax costs of the various sources of invested capital raised by the firm to finance its operations and

Figure 1 Matching Cash Flows and Discount Rates in DCF Analysis

	Object of the Valuation Exercise	
Steps	**Equity Valuation**	**Project or Firm Valuation (Debt Plus Equity Claims)**
Step 1: Estimate the amount and timing of future cash flows.	Equity free cash flow	Project (firm) free cash flow
Step 2: Estimate a risk-appropriate discount rate.	Equity required rate of return	Combine debt and equity discount rate (weighted average cost of capital—WACC).
Step 3: Discount the cash flows.	Calculate the present value of the estimated Equity FCFs using the equity discount rate to estimate the value of the equity invested in the project.	Calculate the present value of the Project or firm FCF using the WACC to estimate the value of the project (firm) as an entity.

investments. We define the firm's **invested capital** as capital raised through the issuance of interest-bearing debt and equity (both preferred and common). Note that the above definition of invested capital specifically excludes all non-interest-bearing liabilities such as accounts payable, as well as unfunded pension liabilities and leases. This is because we will be calculating what is known as the firm's **enterprise value,** which is equal to the sum of the values of the firm's equity and interest-bearing liabilities. Note, however, that these excluded sources of capital (e.g., unfunded pension liabilities and leases) do affect enterprise value, because they affect the firm's future cash flows.

Equation 1 defines the WACC as the average of the estimated required rates of return for the firm's interest-bearing debt (k_d), preferred stock (k_p), and common equity (k_e). The weights used for each source of funds are equal to the proportions in which funds are raised. (That is, w_d is the weight attached to debt, w_p is the weight associated with preferred stock, and w_e is the weight attached to common equity.)

$$\text{WACC} = k_d(1 - \text{T})w_d + k_p w_p + k_e w_e \tag{1}$$

Note that the cost of debt financing is the rate of return required by the firm's creditors, k_d, adjusted downward by a factor equal to 1 minus the corporate tax rate $(1 - \text{T})$ to reflect the fact that the firm's interest expense is tax-deductible. Thus, the creditors receive a return equal to k_d, but the firm experiences a net cost of only $k_d(1 - \text{T})$. Because the cost of preferred and common equity is not tax-deductible, there is no need for a similar tax adjustment to these costs.

The mechanics of calculating a firm's WACC can be summarized with the following three-step procedure:

- **Step 1:** Evaluate the firm's capital structure and determine the relative importance of each component in the mix (i.e., capital structure weights).
- **Step 2:** Estimate the opportunity cost of each of the sources of financing and adjust it for the effects of taxes where appropriate.
- **Step 3:** Finally, calculate the firm's WACC by computing a weighted average of the estimated after-tax costs of capital sources used by the firm (i.e., by substituting into Equation 1).

As you might suspect, there are some estimation issues that arise with respect to both weights and opportunity costs that need to be addressed. Let's consider each in turn.

Use market weights First, with regard to the capital structure weights, it is important that the components used to calculate the WACC formula reflect the current importance of each source of financing to the firm. This means that the weights should be based on the market rather than book values of the firm's securities, because market values, unlike book values, represent the relative values placed on the firm's securities at the time of the analysis (rather than at some previous time when the securities were issued).

Use market-based opportunity costs The second estimation issue that arises in calculating the firm's WACC relates to the rates of return or opportunity costs of each source of capital. Just as was the case with the capital structure weights, these costs should reflect the current required rates of return, rather than historical rates at the time the capital was raised. This reflects the fact that the WACC is an estimate of the firm's opportunity cost of capital today.

Use forward-looking weights and opportunity costs Firms typically update their estimate of the cost of capital periodically (e.g., annually) to reflect changing market conditions. However, in most cases, analysts apply the WACC in a way that assumes that it will be constant for all future periods. This means that they implicitly assume that the weights for each source of financing, the costs of capital for debt and equity, and the corporate tax rate are constant. Although it is reasonable to assume that the components of WACC are constant so long as the firm's financial policies remain fixed, there are circumstances in which financial policies will change in predictable ways over the life of the investment. Leveraged buyout, or LBO, financing typically involves a very high level of debt that is subsequently paid down. In this case, we will find it useful to apply another variant of the DCF model (the adjusted present value model).

Discounted Cash Flow, Firm Value, and the WACC

The connection of the WACC to the discounted cash flow (DCF) estimate of firm value is captured in Equation 2 below:[1]

$$\begin{matrix}\text{Firm}\\ \text{Value}_0\end{matrix} = \sum_{t=1}^{N} \frac{E(\text{Firm FCF}_t)}{(1 + \text{WACC})^t} \tag{2}$$

$E(\text{Firm FCF}_t)$ is the expected free cash flow earned by the firm in period t, where "firm" FCF is analogous to "project" FCF. For our purposes, a project can be considered a mini-firm, and thus a firm is simply the combination of many projects. Consequently, firm and project free cash flows are calculated in exactly the same way.

Equation 2 expresses Firm Value with a "0" subscript to indicate that we are determining firm value *today* ("time zero"), based on cash flows starting one period hence. In general, analysts assume that the period length is one year, and they ignore the fact that the cash flow accrues over the course of the year. We will follow this *end-of-year convention* throughout the book; however, we recognize that many financial analysts assume that cash flows arrive at the middle of the year, to account for the fact that for most projects, cash flows occur throughout the year.

Illustration—Using Discounted Cash Flow Analysis to Value an Acquisition

To illustrate the connection between the WACC and valuation, consider the situation faced by an analyst working for Morgan Stanley, who has a client interested in acquiring OfficeMart Inc. a retailer of office products. This client is interested in purchasing the entire firm, which means that it will acquire all of its outstanding equity and assume its outstanding debt. Although valuations of company acquisitions can be quite involved, the Morgan Stanley analyst has made a simple "first-pass"

[1] Equation 2 does not reflect the value of the firm's nonoperating assets, nor does it capture the value of the firm's excess liquidity (i.e., marketable securities).

analysis of the intrinsic value of OfficeMart, following the three-step discounted cash flow process.

Step 1: *Forecast the amount and timing of free cash flows.* Because we are interested in valuing the firm as an entity (both debt- and equity-holder claims), we estimate firm FCFs in the same way as we have in the past. Thus, for the coming year, the Morgan Stanley analyst estimates that OfficeMart's cash flow will be $560,000, as follows:

Sales	$ 3,000,000
Cost of goods sold	(1,800,000)
Depreciation	(500,000)
Earnings before interest and taxes (EBIT)	700,000
Taxes[2](20%)	(140,000)
Net operating profit after taxes (NOPAT)	560,000
Plus: Depreciation	500,000
Less: Capital expenditures (CAPEX)	(500,000)
Less: Change in net working capital (WC)	(0)
Firm FCF	$ 560,000

Step 2: *Estimate the appropriate discount rate.* Because we are valuing the entire company (debt plus equity claims), the discount rate we choose should represent a combination of the rates that are appropriate for both debt and equity, or the weighted average cost of capital.

OfficeMart finances 40% of its assets using debt costing 5%; equity investors in companies similar to OfficeMart (in terms of both the industry and their capital structures) demand a 14% return on their investment. Combining OfficeMart's after-tax cost of borrowing (interest expense is tax-deductible and the firm's tax rate is 20%) with the estimated cost of equity capital, we calculate a weighted average cost of capital for the firm of 10% [i.e., $5\%(1 - 20\%)0.4 + 14\% \times 0.6 = 10\%$].

Step 3: *Discount the estimated cash flows.* OfficeMart's estimated cash flows form a level perpetuity. That is, each year's cash flow is equal to the prior year's cash flow, or $560,000. Consequently, we can calculate the present value of the firm's future cash flows as follows:

$$\text{Value of Office Mart Inc.} = \frac{\$560{,}000}{.10} = \$5{,}600{,}000$$

Thus, this first-pass analysis suggests that the value of OfficeMart is $5,600,000.

Recall that we have estimated firm value, which is the sum of both the firm's debt and equity claims. If we want to estimate the value of the firm's *equity,* we need to subtract the value of its debt claims from the $5,600,000 valuation of the firm. Because OfficeMart has $2,240,000 worth of debt outstanding, the value of the firm's equity is $5,600,000 − $2,240,000 = $3,360,000.

[2] This is *not* the firm's actual tax liability, because it is based on earnings before interest expense has been deducted. See the Technical Insight box on interest tax savings.

TECHNICAL INSIGHT

What About Interest Tax Savings?

Note that the taxes in the Project FCF and Firm FCF calculations are based on the level of operating profits or earnings *before* interest has been deducted. In other words, we are calculating the after-tax cash flows of an all-equity firm. However, interest expense and its tax deductibility are not ignored in our valuation. We use the *after-tax* cost of debt when calculating the weighted average cost of capital, which accounts for the tax savings associated with the tax deductibility of interest. For example, if the required rate of return for debt is 7% and the firm pays taxes at a rate of 30%, the after-tax cost of debt is 4.9%—i.e., .07(1 − .30) = .049, or 4.9%, while the pretax cost of debt is 7%.

Equity Valuation

In the middle column of Figure 1, we noted that the value of the firm's equity can also be calculated directly by discounting the equity free cash flow, or Equity FCF, back to the present using the equity holder's required rate of return. Because we have not yet defined Equity FCF, let's take a moment to reflect on how Equity FCFs are related to Firm FCFs.

Equity free cash flow To calculate Equity FCF, recall that a firm's FCF is equal to the sum of the cash flows available to be paid to the firm's creditors (creditor cash flows) and owners (Equity FCF), i.e.,

$$\begin{matrix}\text{Firm}\\ \text{FCF}\end{matrix} = \begin{matrix}\text{Creditor}\\ \text{Cash Flows}\end{matrix} + \begin{matrix}\text{Equity}\\ \text{FCF}\end{matrix}$$

such that

$$\begin{matrix}\text{Equity}\\ \text{FCF}\end{matrix} = \begin{matrix}\text{Firm}\\ \text{FCF}\end{matrix} - \begin{matrix}\text{Creditor}\\ \text{Cash Flows}\end{matrix}$$

So, what are the cash flows paid to, or received from, the firm's creditors? There are two types of cash flows to consider: interest and principal. Also, because interest expenses are tax-deductible, we adjust interest to reflect the interest tax savings such that after-tax interest equals the firm's interest expense less the tax savings resulting from the deduction of interest from the firm's taxable income. The principal payments can be both cash inflows when the firm initiates new borrowing and a cash outflow when the firm repays debt. We can summarize the calculation of creditor cash flows as follows:

$$\begin{matrix}\text{Creditor}\\ \text{Cash Flows}\end{matrix} = \underbrace{\begin{matrix}\text{Interest}\\ \text{Expense}\end{matrix} - \begin{matrix}\text{Interest}\\ \text{Tax Savings}\end{matrix}}_{\text{After-Tax Interest Expense}} + \underbrace{\begin{matrix}\text{Principal}\\ \text{Payments}\end{matrix} - \begin{matrix}\text{New Debt Issue}\\ \text{Proceeds}\end{matrix}}_{\begin{matrix}\text{Net Debt Proceeds}\\ \text{(Change in Principal)}\end{matrix}}$$

Thus, we define Equity FCF as follows:

$$\begin{matrix}\text{Equity}\\ \text{FCF}\end{matrix} = \begin{matrix}\text{Firm}\\ \text{FCF}\end{matrix} - \left(\begin{matrix}\text{Interest}\\ \text{Expense}\end{matrix} - \begin{matrix}\text{Interest}\\ \text{Tax Savings}\end{matrix} + \begin{matrix}\text{Principal}\\ \text{Payments}\end{matrix} - \begin{matrix}\text{New Debt Issue}\\ \text{Proceeds}\end{matrix}\right)$$

Or, removing the parentheses from around the creditor cash flows, Equity FCF is calculated as follows:

$$\begin{matrix}\text{Equity} \\ \text{FCF}\end{matrix} = \begin{matrix}\text{Firm} \\ \text{FCF}\end{matrix} - \begin{matrix}\text{Interest} \\ \text{Expense}\end{matrix} + \begin{matrix}\text{Interest} \\ \text{Tax Savings}\end{matrix} - \begin{matrix}\text{Principal} \\ \text{Payments}\end{matrix} + \begin{matrix}\text{New Debt Issue} \\ \text{Proceeds}\end{matrix}$$

Equity valuation example—OfficeMart Inc. Let's return to our OfficeMart example to compute Equity FCF and the value of the firm's equity. Earlier, we learned that OfficeMart has borrowed \$2,240,000 on which it pays 5% interest before deducting the interest from the firm's taxes, which are 20% of its income. This means that the firm will pay \$112,000 in interest expense each year. Assuming the debt is not repaid but remains outstanding forever, then the firm has no principal repayments, and if it borrows no added funds, the creditor cash flows equal \$89,600 = \$112,000 − 112,000 × .20. Given that the firm's FCF is \$560,000, we calculate Equity FCF as follows:

$$\begin{matrix}\text{Equity} \\ \text{FCF}\end{matrix} = \begin{matrix}\text{Firm} \\ \text{FCF}\end{matrix} - \begin{matrix}\text{Interest} \\ \text{Expense}\end{matrix} + \begin{matrix}\text{Interest} \\ \text{Tax Savings}\end{matrix} - \begin{matrix}\text{Principal} \\ \text{Payments}\end{matrix} + \begin{matrix}\text{New Debt Issue} \\ \text{Proceeds}\end{matrix}$$

$$\begin{matrix}\text{Equity} \\ \text{FCF}\end{matrix} = \$560{,}000 - 112{,}000 + 22{,}400 - 0 + 0 = \$470{,}400$$

To value the equity in the project, we discount the Equity FCF stream (note that the Equity FCFs are a level perpetuity—a constant cash flow received annually in perpetuity) using the cost of equity capital, which was earlier assumed to be 14%, i.e.,

$$\begin{matrix}\text{Value of} \\ \text{Office Mart's} \\ \text{Equity}\end{matrix} = \frac{\$470{,}400}{.14} = \$3{,}360{,}000$$

So the value of OfficeMart's equity is equal to the residual value remaining after the firm's debt obligations are deducted from the value of the firm (i.e., \$5,600,000 − \$2,240,000 = \$3,360,000), and it is also equal to the present value of the equity free cash flows! Whether equity value is estimated directly by discounting Equity FCFs or indirectly by subtracting the value of the firm's debt from firm value varies in business practice. For example, it is common practice among private equity investors to value equity directly. However, when evaluating firms in the context of mergers and acquisitions, the valuations generally focus on the firm as a whole.

3 Estimating the WACC

In this section, we introduce techniques for estimating a firm's WACC. In our discussion, we will frequently reference the practices of Ibbotson Associates, which is a prominent source of information (see the Practitioner Insight box on the opposite box). Ibbotson follows the same three-step process we discussed earlier for the estimation of a firm's WACC but applies the procedure to *industry groups*. The rationale for focusing on the industrywide cost of capital is based on the beliefs that (1) the variation in the cost of capital within industries is modest when compared to the variation between industries, and (2) estimation errors are minimized if we focus on groups of similar firms from the same industry.[3]

[3] For a complete discussion of its methodology, see Ibbotson Associates, *Cost of Capital—Yearbook*.

PRACTITIONER INSIGHT

Roger Ibbotson on Business Valuation and the Cost of Capital*

The appraisal of a firm's value using discounted cash flow is a relatively straightforward process. The standard procedure involves estimating the firm's cost of capital and future cash flows and then discounting the cash flows back to the present using the cost of capital. Unfortunately, estimates of cash flows as well as discount rates are fraught with errors. The potential for estimation errors is probably greatest for cash flow estimates because there are no anchors or boundaries that can be used to provide guidance about what reasonable estimates might be. In contrast, in making cost of capital estimates, the analyst has historical market returns on risk-free securities and common stock (in total and by industry), which provide some indication of the neighborhood in which a firm's cost of capital might reside.

At Ibbotson Associates, we provide historical information about stock and bond returns that indicates appropriate required rates of return on investments with different risks. We also provide alternative models that can be used to calculate the cost of capital. Some of the cost of capital models can seem overly complex when conveyed to upper management, board members, or others who lack financial sophistication. Moreover, estimated costs of capital (particularly equity capital) can, and often do, vary widely. My experience suggests that the following guidelines can be useful when addressing the estimation of a company's cost of capital:

- *Use the simpler models when possible.* Our experience suggests that some of the more sophisticated models of the cost of equity have the highest estimation errors.
- *Account for company size and industry.* These variables are unambiguously measured and have been reliably demonstrated to be related to the firm's cost of capital.
- *Calculate the cost of capital using several methods.* Even though you may rely principally on simpler models, each method is informative.

*Roger Ibbotson is founder and former chairman of Ibbotson Associates Inc. Ibbotson Associates is the largest provider of cost of capital information in the United States. Currently, its *Cost of Capital Quarterly* includes industry cost of capital analysis on more than 300 U.S.-based industries for help in performing discounted cash flow analysis. The data include industry betas, costs of equity, weighted average costs of capital, and other important financial statistics presented by industry.

Evaluate the Firm's Capital Structure Weights—Step 1

The first step in our analysis of WACC involves the determination of the weights that are to be used for the components of the firm's capital structure. These weights represent the fraction of the firm's **invested capital** contributed by each of the sources of capital. Note that we need to be a bit careful here when talking about invested capital, for it does not include everything on the right-hand side of the firm's balance sheet. Specifically, a firm's invested capital is the sum of *only* the firm's interest-bearing debt, preferred equity, and common equity.

In theory, we should calculate the weights using observed market prices for each of the firm's securities (be they debt or equity), multiplied by the number of outstanding securities. In practice, however, capital structure weights are typically calculated using market values *only for equity securities* (preferred[4] and common stock). The market

[4] We use the book value of the firm's preferred stock outstanding, as does Ibbotson Associates. However, if preferred share prices are available, the market capitalization for these shares should be used.

prices of equity securities are readily available, so an analyst can simply multiply the current market price of the security by the number of shares outstanding to calculate total market values. For debt securities, book values are often substituted for market values, because market prices for corporate debt are often difficult to obtain (see Industry Insight box). However, when market values of debt are available, they should be used in place of book values.

INDUSTRY INSIGHT

Estimating the Cost of Debt in Practice

Ibbotson Associates estimates the industry cost of debt using the yield curve for various debt ratings based on the Merrill Lynch U.S. Domestic Bond Indices. As a practical matter, bonds are classified into one of three groups (see Figure 2): investment grade (S&P ratings of AAA, AA, A, and BBB), below investment grade (S&P ratings of BB, B, CCC, CC, and D), and not rated. Yields for each group and maturity are then averaged to arrive at an estimate of the yield curve for debt. The average yields for the lower two groups are used as proxies for bonds that are not rated. A weighted average cost of debt can then be calculated for each company in an industry using yields for the firm's debt rating and weights determined by the actual debt maturing in a particular year compared to all debt outstanding. To get an industry cost of debt, the average yields for each company in the industry are averaged. Ibbotson Associates uses debt maturing in each of the next five years, and then assumes that all remaining debt matures in Year 6.

The Cost of Debt—Step 2

In theory, we would like to estimate the expected return that investors require on the firm's debt. In practice, analysts typically use the promised *yield to maturity* on the firm's outstanding debt as their estimate of the expected cost of debt financing.

Promised or Contractual Yields to Maturity on Corporate Bonds

Estimation of the yield to maturity (YTM) on a corporate bond issue is straightforward when the analyst has access to information regarding the maturity of the bond, its current market price, the coupon rate of interest, and the schedule of principal payments. For example, in the fall of 2009, Home Depot (HD) had a bond issue that was due in 2016 (i.e., 7 years to maturity), carried an annual coupon rate of 5.4%, paid semiannual interest and therefore had 14 payments, and had a market price of \$1,049. The YTM on the bond issue can be calculated by solving for YTM in the following bond valuation equation:

$$\$1{,}049 = \frac{(.054/2) \times \$1{,}000}{(1 + \text{YTM})^1} + \frac{(.054/2) \times \$1{,}000}{(1 + \text{YTM})^2} + \frac{(.054/2) \times \$1{,}000}{(1 + \text{YTM})^3} + \cdots + \frac{(.054/2) \times \$1{,}000 + \$1{,}000}{(1 + \text{YTM})^{14}}$$

The semiannual YTM is 2.287%, which converts to an annualized YTM of 5.82%.[5] Based on a marginal tax rate of 35%, the after-tax cost of Home Depot's bond issue is then $3.783\% = 5.82 \times (1 - .35)$.

Calculating the YTM of a firm's debt is difficult for firms with a large amount of debt that is privately held and, hence, does not have market prices that are readily available. Because of this, it is standard practice to estimate the cost of debt using the yield to

[5] Annual YTM $= (1 + \text{Semiannual YTM})^2 - 1 = (1 + .0291)^2 - 1 = .059$ or 5.9%. This quantity is often referred to as the effective annual rate, or just yield.

Figure 2 **A Guide to Corporate Bond Ratings**

Moody's	S&P	Fitch	Definitions
Aaa	AAA	AAA	Prime (maximum safety)
Aa1	AA+	AA+	High grade, high quality
Aa2	AA	AA	
Aa3	AA–	AA–	
A1	A+	A+	Upper medium grade
A2	A	A	
A3	A–	A–	
Baa1	BBB+	BBB+	Lower medium grade
Baa2	BBB	BBB	
Baa3	BBB–	BBB–	
Ba1	BB+	BB+	Noninvestment grade
Ba2	BB	BB	Speculative
Ba3	BB–	BB–	
B1	B+	B+	Highly speculative
B2	B	B	
B3	B–	B–	
Caa1	CCC+	CCC	Substantial risk
Caa2	CCC	—	In poor standing
Caa3	CCC–	—	
Ca	—	—	Extremely speculative
C	—	—	May be in default
—	—	DDD	Default
—	—	DD	
—	D	D	

maturity on a portfolio of bonds with similar credit ratings and maturity as the firm's outstanding debt.

Reuters provides average spreads to Treasury data that are updated daily and cross-categorized by both default rating and years to maturity, as found in Figure 3. Using our Home Depot bond issue, we can estimate the YTM for a 7-year bond with a default rating of Aa3/AA–. Such a bond has a spread to the 7-year Treasury bond YTM of .52%, or 52 basis points (see boxed number in Figure 3). Given that the 7-year Treasury YTM was 3.00% at the time of this writing, this provides an estimate of 3.00% + .52% = 3.52% for the YTM on Home Depot's bonds.

Promised Versus Expected Rates of Return

The yield to maturity is calculated using the promised interest and principal payments and thus can be considered a reasonable estimate of the cost of debt financing *only* when the risk of default is so low that promised cash flows are reasonable estimates of expected cash flow. For lower-rated debt, however, promised and expected cash flows are not the same, and an explicit adjustment for the prospect of loss in the event of default becomes necessary.

Figure 3 **Reuters Corporate Spreads (in basis points) for Industrials in 2004***

Rating	1 yr	2 yr	3 yr	5 yr	7 yr	10 yr	30 yr
Aaa/AAA	5	10	15	20	25	33	60
Aa1/AA+	10	15	20	30	35	42	66
Aa2/AA	15	25	30	35	44	52	71
Aa3/AA-	20	30	35	45	52	59	78
A1/A+	25	35	40	50	55	65	85
A2/A	35	44	55	60	65	73	90
A3/A-	45	59	68	75	80	89	110
Baa1/BBB+	55	65	80	90	94	104	123
Baa2/BBB	60	75	100	105	112	122	143
Baa3/BBB-	75	90	110	115	124	140	173
Ba1/BB+	115	125	140	170	180	210	235
Ba2/BB	140	180	210	205	210	250	300
Ba3/BB-	165	200	230	235	235	270	320
B1/B+	190	215	250	250	275	335	360
B2/B	215	220	260	300	315	350	450
B3/B-	265	310	350	400	435	480	525
Caa/CCC	1125	1225	1250	1200	1200	1275	1400

*Methodology—Reuters Pricing Service (RPS) has eight experienced evaluators responsible for pricing approximately 20,000 investment-grade corporate bonds. Corporate bonds are segregated into four industry sectors: industrial, financial, transports, and utilities. RPS prices corporate bonds at a spread above an underlying Treasury issue. The evaluators obtain the spreads from brokers and traders at various firms. A generic spread for each sector is created using input from street contacts and the evaluator's expertise. A matrix is then developed based on sector, rating, and maturity.

To elaborate on the distinction between promised yields and expected returns on debt, consider the calculation of the yield to maturity (YTM) on a bond with one year to maturity, found in Equation 3a below:

$$\text{Bond Price} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})} \tag{3a}$$

Note that the cash flows used to determine YTM are the contractual or promised cash payments, which equal expected cash flows *only* for default-free bonds. For debt with default risk, the expected cash flows must reflect the probability of default (Pb) and the recovery rate (Re) on the debt in the event of default. For our one-period bond issue, the cost of debt k_d can be calculated using expected cash flows to the bondholder as follows:

$$\text{Bond Price} = \frac{[\text{Interest} + \text{Principal}] \times (1 - Pb) + [\text{Interest} + \text{Principal}] \times Pb \times Re}{(1 + k_d)} \tag{3b}$$

The expected cash flow to the bondholder is the promised principal and interest payments, weighted by the probability that the debt defaults $(1 - Pb)$, plus the cash flow that is received in the event of default, weighted by the probability of default (Pb). To illustrate this, consider a bond with a \$1,000 face value and one year to maturity that is currently selling for \$985.00 and paying a 9% interest payment annually. Using Equation 3a, we estimate the bond's YTM to be 10.66%, i.e.,

$$\begin{array}{c}\text{Bond}\\ \text{Price}\end{array} = \frac{\text{Interest} + \text{Principal}}{(1 + \text{YTM})^1} = \frac{.09 \times \$1{,}000 + \$1{,}000}{(1 + .1066)} = \$985$$

The 10.66% YTM represents the rate of return an investor would realize if the bond does not default. If, however, the probability of default on the bond is 15% and the recovery rate is 75%, the expected rate of return to the investor, using Equation 3b, is 6.51%.

In practice, the differences between YTM and k_d are relatively modest for investment-grade debt (i.e., debt rated BBB or higher), and the YTM provides a reasonable estimate of the cost of debt. However, for firms with below-investment-grade debt, there can be a meaningful difference between the *promised yield to maturity* on the debt and the *expected return* or cost of the debt (as we demonstrated in the above example).

There are two ways to estimate the cost of debt for below-investment-grade debt. The first method involves estimating the expected cash flows of the debt using expected rates of default and recovery rates, and using these expected cash flows to calculate the internal rate of return on the debt. We illustrate how this can be done in the Appendix. The second method applies the capital asset pricing model (CAPM), which we discuss in more detail later in the chapter when we discuss the cost of equity capital. Very briefly, the CAPM requires an estimate of the beta of the debt, along with an expected return premium on the stock market. For example, betas of low-rated bonds are approximately .4, while bonds with AAA ratings generally have betas that are about .2. If we assume a market risk premium of 5%, then the spread between the expected returns of a AAA bond and a below-investment-grade bond is approximately $(0.4 - 0.1)$ times 5%, which equals 1.5%. Therefore, if the current yield and expected return on a AAA bond (our proxy for the risk-free rate) is 6%, then the expected return on the lower-rated bond is about 7.5%.

Estimating the Cost of Convertible Corporate Bonds

Convertible bonds represent a hybrid form of financing that is both debt and equity because the holder of the bond can, at his or her discretion, convert the bond into a prescribed number of common shares. Because these bonds have the conversion feature, they typically carry a lower rate of interest and consequently their estimated yield to maturity understates the true cost of debt. This "dual" source of value means that the cost of financing by issuing convertibles is a function of both the underlying security (bond or preferred stock) and the call option. Consequently, the cost of capital obtained by issuing convertible bonds can be thought of as a weighted average of the cost of issuing straight bonds and the cost of the exchange feature (call option), where the weights equal the relative contributions of the two components to the value of the security. We delve into the estimation of the cost of convertible debt in the Appendix.

The Cost of Preferred Equity—Step 2 (continued)

Estimating the cost of straight (i.e., nonconvertible) preferred stock is straightforward, because it typically pays the holder a fixed dividend each period (quarterly), forever.[6] The value of such a stream of dividends can be found as follows:

$$\begin{array}{c}\text{Preferred}\\ \text{Stock}\\ \text{Price } (P_{ps})\end{array} = \frac{\text{Preferred Dividend } (Div_{ps})}{\text{Required Return } (k_{ps})} \tag{4a}$$

Using the preferred dividend and observed price of preferred stock, we can infer the investor's required rate of return as follows:

$$k_{ps} = \frac{Div_{ps}}{P_{ps}} \tag{4b}$$

To illustrate, consider the preferred shares issued by Alabama Power Company (ALP-PP), which pay a 5.3% annual dividend on a $25.00 par value, or $1.33 per share. On November 4, 2009, these preferred shares were selling for $23.60 per share. Consequently, investors require a 5.67% return on these shares, calculated as follows:

$$k_{ps} = \frac{\$1.33}{23.60} = .564 \text{ or } 5.64\%$$

Note that the preferred dividend is also a *promised* dividend, in the same way that the interest on corporate bonds is *promised* interest, and thus *does not* necessarily represent the dividend that the preferred stockholder expects to receive. This means that the 5.64% return calculated above provides an upper limit on the cost of preferred stock, because the firm may decide to suspend payment or become bankrupt.[7]

The key thing to remember here is that the standard method for estimating the cost of preferred stock using Equation 4a is biased upward, yielding estimated costs that are higher than the expected costs. However, it is standard practice to use the promised dividend divided by the price of the preferred stock, as we did in Equation 4b.

The Cost of Common Equity—Step 2 (continued)

The cost of common equity capital (k_e) is the most difficult estimate we have to make in evaluating a firm's cost of capital. The difficulty arises from the fact that the common shareholders are the residual claimants of the firm's earnings. That is, the common stockholders receive a return out of what is left over after all other claimants (bondholders and preferred stockholders) have been paid. Hence, there is no promised or prespecified return based on a financial contract (as is the case with both bondholders and preferred stockholders).

[6] We discuss a method for evaluating the conversion feature with respect to corporate bonds in the Appendix. A similar approach can be taken for the evaluation of convertible preferred shares.

[7] In contrast to corporate bond interest and principal payments, the issuing firm can suspend payment of the preferred dividend without being forced into bankruptcy. This makes the difference in promised and expected dividends even more dramatic than with bonds, because the issuing firm is less likely to default on interest and principal payments.

The relevant cost of equity is simply the rate of return investors *expect* from investing in the firm's stock. This return comes in the form of cash distributions (i.e., dividends and cash proceeds from the sale of the stock). We review two broad approaches that are each widely used to estimate the cost of equity. The first consists of what financial economists refer to as *asset pricing models.* Specifically, we present three variants of the *Capital Asset Pricing Model (CAPM).*

The second approach has a much older heritage in finance and comes out of the pioneering work of John Burr Williams (1938) and, later, Myron Gordon (1962).[8] This approach, which is often referred to as the *discounted cash flow approach,* first estimates the expected stream of dividends and then calculates the implied cost of equity capital, or equivalently the internal rate of return, that makes the present value of the dividend stream equal the firm's stock price. This implied return is then used as the firm's cost of equity.

Method 1—Asset Pricing Models

Asset pricing theories entered the finance literature in the 1960s with the introduction of the **capital asset pricing model,** or **CAPM.** The traditional CAPM was followed by a host of modified versions that relaxed some of the more stringent assumptions upon which the original theory was based. We consider three versions: the traditional CAPM, the size-adjusted CAPM, and multifactor models.

The traditional CAPM

One of the basic tenets of finance is that if investors are risk-averse, they will require a higher rate of return to hold riskier investments. The important question addressed by the CAPM is how one should measure risk. The basic intuition of the CAPM is that the relevant risk of a stock is determined by how that stock contributes to the overall volatility of a well-diversified portfolio. As it turns out, there are some stocks that are quite volatile; that is, their returns vary a lot from month to month, but despite their volatility, they contribute very little to the volatility of a well-diversified portfolio. According to the CAPM, these stocks should require lower rates of return than their counterparts that may be less volatile but that contribute more to the volatility of well-diversified portfolios.

To understand the relation between risk and return, it is useful to decompose the risk associated with an investment into two components. The first component consists of variability that contributes to the risk of a diversified portfolio, and the second component consists of variability that does not contribute to the risk of a diversified portfolio. The first component is generally referred to as either **systematic risk** or **nondiversifiable risk,** and the second component is generally referred to as **nonsystematic risk** or **diversifiable risk.** Sources of systematic risk include market factors such as changes in interest rates and energy prices that influence almost all stocks. The logic of the CAPM suggests that stocks that are very sensitive to these sources of risk should have high required rates of return, because these stocks contribute more to the variability of diversified portfolios. Sources of nonsystematic risk include random firm-specific events such as lawsuits, product defects, and various technical innovations. The logic of the CAPM suggests that these sources of risk should have almost no effect on required rates of return because they contribute very little to the overall variability of diversified portfolios.

[8] J. B. Williams, *Theory of Investment Value* (Cambridge, MA: Harvard University Press, 1938), and M. Gordon, *The Investment, Financing, and Valuation of the Corporation* (Homewood, IL: Irwin, 1962).

The CAPM can be expressed as the following equation, which relates the required expected return of an investment to systematic risk:

$$k_e = k_{rf} + \beta_e(k_m - k_{rf}) \tag{5}$$

where

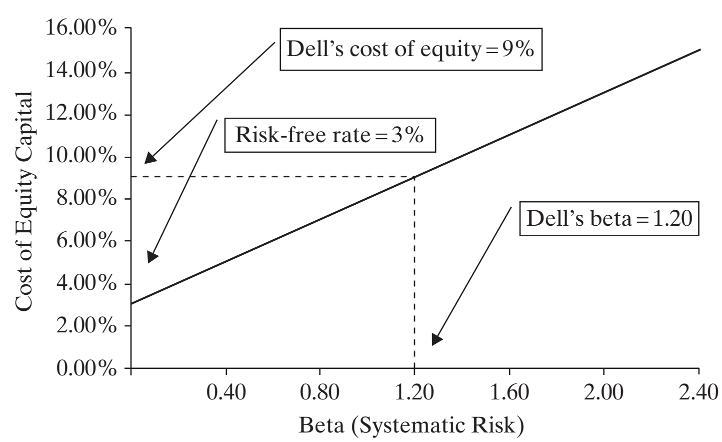
k_{rf} = the risk-free rate of interest
β_e = the beta, or systematic risk of the company's common equity, which is estimated from a regression of a stock's return minus the risk-free rate on a market return, such as the S&P 500 return, minus the risk-free rate
k_m = the expected return on the overall market portfolio comprised of all risky assets
$(k_m - k_{rf})$ = the expected equity risk premium (the expected return on the overall market minus the risk-free rate)

Figure 4 illustrates the connection between systematic risk and the expected rate of return on common equity. For example, to determine the cost of equity to Dell Corporation, let's assume that we estimate its beta coefficient to be 1.20, the risk-free rate of interest to be 3.00%, and the market risk premium to be approximately 5%.[9] We can substitute into Equation 5 to estimate the expected cost of equity capital for the firm as 9.00%.

We now turn to a more in-depth discussion of the three CAPM inputs: the risk-free interest rate, beta, and the market risk premium.

Selecting the risk-free rate of interest The risk-free rate of interest is the least controversial estimate we have to make for the CAPM inputs. Nevertheless, there are two issues we need to address: First, what is a risk-free security, and second, what maturity should we use? Unfortunately, the CAPM provides little guidance in responding to either of these issues.

Figure 4 **Estimating Dell Corporation's Cost of Equity Using the CAPM**



[9] We discuss the market risk premium in detail later in the chapter.

Identifying the risk-free rate Analysts typically use current yields on U.S. Treasury securities to define the risk-free rate of interest when evaluating the cost of capital in the United States.[10] When applying the CAPM in other economies, it is customary to use their domestic risk-free rates so as to capture differences in rates of inflation between the United States and that economy.

Choosing a maturity As a general rule, we want to match the maturity of the risk-free rate with the maturity of the cash flows being discounted. In practice, maturity matching is seldom done, however. Most textbooks suggest that short-term rates be used as the risk-free rate, because they are consistent with the simplest version of the CAPM. However, because the estimated cost of equity is typically used to discount distant cash flows, it is common practice to use a long-term rate for, say, 10- or 20-year maturities as the risk-free rate.[11] We agree with this practice; however, as we note below, the beta estimate used in the CAPM equation should also reflect this longer-maturity risk-free rate.

Estimating the beta The firm's **beta** represents the sensitivity of its equity returns to variations in the rates of return on the overall market portfolio. That is, if the value of the market portfolio of risky investments outperforms Treasury bonds by 10% during a particular month, then a stock with a beta coefficient of 1.25 would be expected to outperform Treasury bonds by 12.5%. A stock's beta should be estimated by regressing the firm's excess stock returns on the excess returns of a market portfolio, where excess returns are defined as the returns in excess of the risk-free return, as shown in the following equation:

$$(k_e - k_{rf})_t = \alpha + \beta_e(k_m - k_{rf})_t + e_t \tag{6}$$

where

k_e = the observed rate of return earned for investing in the firm's equity in period t,
k_{rf} = the risk-free rate of interest observed in period t,
α = a constant (intercept) term,
β_e = the beta for the company's common equity,
$(k_m - k_{rf})$ = the equity risk premium, and
e_t = the error term (that part of the equity return that is not explained by overall market movements)

Note that many analysts make a common mistake. They fail to match the maturity of the risk-free rate, used in Equation 6 to calculate beta, with the maturity of the rate used to calculate the equity risk premium. Very simply, if you use a long-term Treasury bond yield as the risk-free rate, then the excess return on the market used to calculate beta should be the excess return of the market portfolio over the long-term Treasury bond return.[12]

[10] Because Treasury securities have special appeal to foreign central banks and are exempt from state taxes, we cannot expect a zero-beta stock to have a return as low as the yield on a Treasury security. Because of this, one might want to use the AAA corporate bond or equivalent commercial paper rate instead of the Treasury rate. However, we follow industry practice and use the long-term Treasury rate.

[11] For example, Ibbotson Associates uses the long-term government bond yield to approximate the risk-free rate in its cost of capital calculations.

[12] It is our understanding that most publicly available betas are estimated with short-term risk-free rates or simply by regressing stock returns on market returns.

Technical Insight

Equity Betas and Financial Leverage

When using the CAPM to estimate the cost of equity, it is standard practice to estimate the firm's equity beta by computing averages of equity betas of comparable firms with similar operating risk characteristics. Because equity betas vary with the use of financial leverage, we need to adjust the betas of the comparison firms for differences in how the firms are financed. This requires an understanding of the relationship between financial leverage and beta.

To develop an understanding of this relationship, we begin by noting that the beta for a firm is simply a weighted average of the betas of each of the sources of capital in the firm's capital structure. So the beta for firm j can be defined as follows:

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j}\left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j}\right) + \beta_{\text{Debt}_j}\left(\frac{\text{Debt Value}_j}{\text{Firm Value}_j}\right) \qquad \text{(i)}$$

Because Firm Value = Equity Value + Debt Value, we can solve for the equity beta by solving equation (i) for β_{Equity_j} as follows:

$$\beta_{\text{Equity}_j} = \beta_{\text{Firm}_j}\left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right) - \beta_{\text{Debt}_j}\left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right) \qquad \text{(ii)}$$

Note that equation (ii) does not explicitly account for the effects of corporate taxes. When corporate taxes favor debt financing, firm values will be affected by the level of debt, and this will in turn affect the beta of the firm as well as the beta of its equity. For example, if corporate income is subject to income taxes at a rate T_c and interest is tax-deductible, the value of the firm is equal to the sum of the hypothetical value of the firm, assuming that it is unlevered (if it has no debt), plus the value of the interest tax savings resulting from the use of debt in the firm's capital structure, i.e.,[13]

$$\begin{matrix}\text{Firm} \\ \text{Value}_j\end{matrix} = \begin{matrix}\text{Debt} \\ \text{Value}_j\end{matrix} + \begin{matrix}\text{Equity} \\ \text{Value}_j\end{matrix} = \begin{matrix}\text{Firm} \\ \text{Value}_{\text{Unlevered}_j}\end{matrix} + \begin{matrix}\text{Value of} \\ \text{the Interest} \\ \text{Tax Savings}_j\end{matrix} \qquad \text{(iii)}$$

The firm's beta can now be expressed in terms of the beta of the unlevered firm and the firm's interest tax savings:

$$\beta_{\text{Firm}_j} = \beta_{\text{Unlevered Firm}_j}\left(\frac{\text{Firm Value}_{\text{Unlevered}_j}}{\text{Firm Value}_j}\right) + \beta_{\text{Interest Tax Savings}_j}\left(\frac{\text{Value of Interest Tax Savings}_j}{\text{Firm Value}_j}\right) \qquad \text{(iv)}$$

[13] This result is from Modigliani and Miller (1963).

An important component of the above equation is $\beta_{\text{Interest Tax Savings}_j}$, which is affected by how the firm changes its capital structure over time. We will consider first the case where the level of debt used by the firm is fixed or constant, which implies that the interest tax savings are the same for all future years. We will then consider the case where the debt to value ratio is fixed, which implies that the interest tax savings vary from year to year in direct relation to changes in the value of the firm.

Case 1: Constant Level of Debt

When the amount of debt used by the firm is constant and risk free, then $\beta_{\text{Debt}_j} = 0$ and $\beta_{\text{Interest Tax Savings}_j}$, which is simply T_c times the firm's interest expense, is also risk free and thus that it too has a beta equal to zero. Substituting the zero betas for debt and the interest tax savings into equations (i) and (iv), we get the following result:

$$\beta_{\text{Firm}_j} = \beta_{\text{Equity}_j}\left(\frac{\text{Equity Value}_j}{\text{Firm Value}_j}\right) = \beta_{\text{Unlevered Firm}_j}\left(\frac{\text{Firm Value}_{\text{Unlevered}_j}}{\text{Firm Value}_j}\right)$$

Solving equation (iii) for the value of the unlevered firm and substituting it into the above equation, we find the following relation between the beta for the equity of the levered firm and the beta for the unlevered firm:

$$\beta_{\text{Equity}_j} = \left[1 + (1 - T_c)\frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right]\beta_{\text{Unlevered Firm}_j} \tag{v}$$

Note that equation (v) is appropriate only for the special case where the level of debt financing is fixed and the tax savings from the interest deduction is risk free. Although these are strong assumptions, equation (v) is widely used in practice to describe the relationship between levered and unlevered equity betas.

Case 2: Constant Debt to Value Ratio

An alternative case, which we will encounter throughout the book, assume that the firm's ratio of debt to firm value, rather than the level of debt, is fixed. In other words, if the value of the firm increases, the firm increases its debt level so that its debt to value ratio remains constant. As a result, the firm's debt level (and consequently its interest tax savings) is perfectly correlated with the value of the firm. In this case, the risk of the interest tax savings is the same as the risk of the firm, implying that the beta of the firm's interest tax savings is equal to β_{Firm_j}.[14]

Substituting $\beta_{\text{Interest Tax Savings}_j} = \beta_{\text{Firm}_j}$ into equation (iv) results in $\beta_{\text{Firm}_j} = \beta_{\text{Unlevered Firm}_j}$. What this means is that the beta of the total firm is unaffected by how the firm is financed, which means that the beta of equity can be solved using equation (ii). Thus, in Case 2, we get the following relationship between the levered and unlevered betas of a firm:

$$\beta_{\text{Equity}_j} = \beta_{\text{Unlevered}_j}\left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right) - \beta_{\text{Dept}_j}\left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right) \tag{vi}$$

[14] This insight is due to Miles and Ezzell (1985).

Solving for the unlevered beta, we get the following:

$$\beta_{\text{Unlevered}_j} = \frac{\beta_{\text{Equity}_j} + \beta_{\text{Debt}_j}\left(\dfrac{\text{Debt Value}_j}{\text{Equity Value}_j}\right)}{1 + \left(\dfrac{\text{Debt Value}_j}{\text{Equity Value}_j}\right)} \qquad \text{(vii)}$$

Case 1 and Case 2 can be viewed as two extremes. In the first, the debt level is never updated, and in the second, capital structure is updated continuously. A more realistic case is one where the firm makes these adjustments annually. When this is the case, the relationship between the levered equity beta and the unlevered firm's beta will be the following:

$$\beta_{\text{Unlevered}_j} = \beta_{\text{Equity}_j}\left(\frac{1}{1 + \left(1 - T_c \times \dfrac{r_d}{1 + r_d}\right)\dfrac{\text{Debt}_j}{\text{Equity}_j}}\right) + \beta_{\text{Debt}_j}\left(\frac{\left(1 - T_c \times \dfrac{r_d}{1 + r_d}\right)\dfrac{\text{Debt}_j}{\text{Equity}_j}}{1 + \left(1 - T_c \times \dfrac{r_d}{1 + r_d}\right)\dfrac{\text{Debt}_j}{\text{Equity}_j}}\right) \qquad \text{(viii)}$$

Note that the only difference in the beta relationships found in equations (vii) and (viii) is the presence of the product of the tax rate and the ratio of the annual rate of interest on the firm's borrowing, r_d, divided by 1 plus this rate, i.e., $T_c \times \frac{r_d}{1 + r_d}$. Equation (viii) is the appropriate relationship between the firm's equity beta with debt financing and the unlevered equity beta if the firm is assumed to update or readjust its capital structure annually. Equation (vii), on the other hand, is the appropriate relationship if updating is continuous and the time between capital structure updates approaches zero and the ratio $\frac{r_d}{1 + r_d}$ also approaches zero because r_d is the rate of interest applicable to the time between capital structure adjustments.

In our calculations in the current chapter, we will assume that the firm maintains a constant debt to value ratio and that it makes adjustments to its debt continuously. Consequently, we unlever beta estimates using equation (vii).

Although we typically estimate a company's beta using historical returns, we should always be mindful that our objective is to estimate the beta coefficient that reflects the relationship between risk and return in the future. Unfortunately, the beta estimate is just that—an estimate—and is subject to estimation error. Fortunately, there are a number of ways to address estimation errors (see the Industry Insight box two pages ahead on alternative beta estimation methods).

The most common fix involves using an average of a sample of beta estimates for similar companies, which has the effect of reducing the influence of random estimation errors. However, it is not enough just to select similar firms, say, from the same industry. Beta coefficients vary not only by industry (or with business risk) but also by the firm's capital structure. Firms that use more financial leverage have higher betas. Consequently, computing a beta from a sample of similar firms' betas is a multistep process. First, we must identify a sample of firms that face similar business risk (usually from the same industry). For example, in Table 1, we use four firms from the drug industry to calculate a beta estimate for Pfizer. Second, we must "unlever" the betas for each of the sample firms to remove the influence of their particular capital structures on their beta coefficients. The relationship between levered and unlevered beta coefficients is defined in Equation 7 in Table 1. Third, we average the unlevered beta coefficients and finally relever them to reflect the capital structure of the firm in question.

In Table 1, the average unlevered beta for Pfizer and the other drug firms is .5723. If we relever this average beta (Step 3) using Pfizer's debt-to-equity capitalization ratio of 25.58%, we get a levered beta estimate for Pfizer of .6419. In this calculation, we place the same weight on Pfizer that we do on each of the other firms. However, because Pfizer is obviously the best match for Pfizer, we may want to increase the weight assigned to Pfizer, say, to be equal to that of all the other firms. If we use this weighting scheme, we get an unlevered beta estimate of .6183, which produces a levered beta estimate of .701.

Estimating the market risk premium Determining the market risk premium requires a prediction of the future spread between the rate of return on the market portfolio and the risk-free return. Remember that the discount rate must reflect the opportunity cost of capital, which is in turn determined by the rate of return associated with investing in other risky investments. If one believes that the stock market will generate high returns over the next 10 years, then the required rate of return on a firm's stock will also be quite high.

Common Valuation Mistakes: Failure to Risk-Adjust Comps

It is common practice when valuing a firm that is privately owned (i.e., its common stock is not publicly traded) to estimate the firm's beta using the observed betas of comparable firms that are publicly traded. However, for a variety of reasons, the **comparison firms** (or *comps*) may have debt ratios that are very different than the debt ratio of the private company that is being valued. When this happens, the analyst can adjust the observed beta of the comparison firm for differences in the debt ratio by using Equation 7.

In the preceding analysis, we made the implicit assumption that the only difference between the private firm being valued and the publicly traded comps related to their use of financial leverage. However, this is not always the case, and analysts often fail to ask why the comparison firm has a different debt ratio than that of the firm being valued. Very simply, the comparison firms may have a different debt ratio because their business risk is more or less than the risk of the firm being valued. For example, small independent refineries can take on less debt than large public refineries like Valero Energy Corp. (VLO), because smaller refineries generally have higher costs of operations and face greater product market risks than their large, publicly traded counterparts.

Table 1 **Estimating the Beta Coefficient for Pfizer Drug Using a Sample of Comparable Drug Firms—Unlevering and Levering Beta Coefficients**

Step 1: Identify firms that operate in the same line of business as the subject firm (i.e., Pfizer). For each, either estimate directly or locate published estimates of its levered equity beta, β_{Equity}, the book value of the firm's interest-bearing debt, and the market capitalization of the firm's equity.[15]

Company Name	Levered Equity Beta	Debt/Equity Capitalization	Assumed Debt Betas	Unlevered Equity Betas
Abbott Laboratories (ABT)	0.2600	18.39%	0.30	0.2662
Johnson & Johnson (JNJ)	0.6300	6.47%	0.30	0.6099
Merck (MRK)	0.9000	10.98%	0.30	0.8407
Pfizer (PFE)	0.7600	25.58%	0.30	0.6663
		15.35%	Average	0.5723

Step 2: Unlever the equity beta coefficient for each firm to get an estimate of the beta for each of the sample firms (including Pfizer) as if it used no financial leverage (i.e., $\beta_{Unlevered}$), and calculate the average for the five unlevered betas.

Industry Insight

Alternative Beta Estimation Methods

The problems we noted with the estimation of beta coefficients from historical return data have led to the development of alternative prediction methods for betas. We will present two of those methods that have gained widespread usage: the BARRA model and the Bloomberg model.

The *BARRA method* is named for its founder, University of California professor of finance Barr Rosenberg. The basic thesis underlying BARRA's methodology is that a stock's beta is not stationary over time, but varies with changes in the fundamental attributes of the firm. Consequently, BARRA developed a forward-looking beta that accounts for current firm characteristics in its beta estimates.

The method is based on research showing that betas estimated from historical returns alone were not as good at predicting future betas as betas that also incorporated consideration for fundamental variables such as an industry variable and various company descriptors. The BARRA method uses an industry variable because it is well known that some industries

[15] Typically, debt is estimated using the book value of the firm's interest-bearing liabilities (short-term notes payable, the current portion of the firm's long-term debt, plus long-term debt). Although we should technically use the market value of the firm's debt, it is customary to use book values, because most corporate debt is thinly traded, if at all. The equity capitalization of the firm is estimated using the current market price of the firm's shares multiplied by the number of outstanding shares. Betas for individual stocks can be obtained from a number of published sources, including virtually all online investment-information sources such as Yahoo Finance or Microsoft's MoneyCentral Web site.

Table 1 *continued*

We unlever the equity beta coefficient using the following relationship between levered (β_{Equity_j}) and unlevered ($\beta_{\text{Unlevered}_j}$) beta coefficients:[16]

$$\beta_{\text{Unlevered}_j} = \frac{\beta_{\text{Equity}_j} + \beta_{\text{Debt}_j}\left(\dfrac{\text{Debt Value}_j}{\text{Equity Value}_j}\right)}{1 + \left(\dfrac{\text{Debt Value}_j}{\text{Equity Value}_j}\right)} \tag{7}$$

where Equity Value$_j$ is equal to the current stock price times the number of outstanding shares of stock. We assume a debt beta of .30 for all firms.[17]

In the preceding example, the average unlevered beta is .4981.

Step 3: Relever the average unlevered equity beta to reflect the target firm's debt-to-equity capitalization ratio and corporate tax rate.

The process of relevering equity betas is simply the reverse of the unlevering process. Technically, we solve Equation 7 for β_{Equity} as follows:[18]

$$\beta_{\text{Equity}_j} = \beta_{\text{Unlevered}_j}\left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right) - \beta_{\text{Debt}_j}\left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j}\right)$$

then substitute the average unlevered beta of .4981 and Pfizer's debt-to-equity capitalization ratio of 6.58% and solve for $\beta_{\text{Levered}} = .5061$.

(such as agriculture and regulated utilities) are persistently low-beta industries, while others (such as electronics, air transportation, and securities firms) are persistently high-beta industries. Moreover, income statement and balance sheet variables are included in the model as predictors. For example, high dividend payout is predictive of low-beta firms, while high covariability of firm earnings with economy-wide earnings is indicative of higher betas.

A second alternative beta estimation model is used by the Bloomberg investment data company. The Bloomberg model adjusts betas estimated using historical data to account for the tendency of historical betas to regress toward the mean. To adjust for this tendency, the Bloomberg adjusted beta uses the following adjustment:

$$\begin{matrix}\text{Bloomberg}\\ \text{Adjusted Beta}\end{matrix} = .33 + .67\left(\begin{matrix}\text{Unadjusted}\\ \text{Historical Beta}\end{matrix}\right)$$

The coefficients of the above model are estimated using past values of contemporaneously estimated betas.

[16] It should be noted that analysts often apply Equation 7 under the assumption that the debt beta is zero.

[17] The beta on debt is in fact close to zero in cases where default is not too likely and when the firm's debt has approximately the same maturity as the maturity of the risk-free debt used to calculate betas. Because most analysts tend to use excess returns using short-term debt to calculate betas, the relevant betas of corporate bonds are between .2 and .4, depending on their default rating and maturity.

[18] If the corporate bond beta is assumed to be zero (as many analysts assume), then the last term on the right-hand side of this expression drops out.

TECHNICAL INSIGHT

Geometric Versus Arithmetic Mean

Consider the following investment in the stock of Carebare Inc. On December 31, 2009, the firm's shares were trading for \$100 a share. One year later, the shares had dropped to only \$50. However, in 2011, the firm experienced a banner year that doubled the share price, leaving it at \$100 on January 1, 2011. If you purchased the shares of Carebare on January 1, 2009, what rate of return did you realize when you sold them on January 1, 2011?

The obvious answer (not considering transaction costs) is 0%, because you ended up exactly where you started. That is,

$$\begin{aligned}\text{Stock Price}_{2009} &= \text{Stock Price}_{2009}(1 + \text{HPR}_{2010})(1 + \text{HPR}_{2011}) \\ &= \text{Stock Price}_{2009}(1 + \text{HPR}_{\text{Mean}})^2\end{aligned}$$

Thus, the mean annual holding period return (HPR) for the two-year period is found by solving for HPR_{Mean}:

$$\text{Stock Price}_{2009}(1 + \text{HPR}_{2009})(1 + \text{HPR}_{2010}) = \text{Stock Price}_{2009}(1 + \text{HPR}_{\text{Mean}})^2$$

or

$$\text{HPR}_{\text{Mean}} = [(1 + \text{HPR}_{2010})(1 + \text{HPR}_{2011})]^{1/2} - 1$$

It turns out that HPR_{Mean} is the *geometric mean* of the HPRs. For example, we can calculate this return by estimating the geometric mean of the returns earned in 2010 and 2011 as follows:

$$\begin{matrix}\text{Geometric} \\ \text{Mean}\end{matrix} = [(1 + -.50)(1 + 1.00)]^{1/2} - 1 = 0\%$$

On the other hand, the *arithmetic mean* of the HPRs for 2009 and 2010 is calculated as follows:

$$\begin{matrix}\text{Arithmetic} \\ \text{Mean}\end{matrix} = \frac{-50\% + 100\%}{2} = 25\%$$

When there are important differences in the business risks of the subject firm and the comps used to estimate its beta, the adjustment for financial leverage in Equation 7 will be incomplete, because it adjusts only for financial risk and not business risk. To address this problem, the analyst must spend lots of effort in selecting the best possible matching firms to use as comps.

Historical Estimates There is no getting around the fact that the market risk premiums that are used in applications of the CAPM are simply guesses. However, they should be educated guesses that are based on sound reasoning.

The approach taken by many analysts is to use past history as a guide to estimate the future market return premium. Figure 5 contains summary statistics on historical

The difference in the geometric and arithmetic mean is generally not as stark as in this example, but it serves to emphasize the points we want to make. The geometric mean return is the appropriate way to measure the rate of return earned during a particular historical return sequence. However, the geometric mean is not the best estimator of future returns unless we expect this sample path to be repeated in the future.

When all future sample paths are equally likely, the best estimate of the forward-looking returns is the arithmetic mean. Technically, we are assuming that each HPR is an independent observation from a stationary underlying probability distribution. So, if we observe annual returns of 10%, −5%, 25%, and 20%, and these annual rates of return are assumed to represent random draws from a single underlying distribution of annual returns, then the best estimate of the mean of this distribution is the arithmetic mean, i.e.,

$$\begin{array}{c}\text{Arithmetic}\\ \text{Mean}\end{array} = \frac{10\% - 5\% + 25\% + 20\%}{4} = 12.5\%$$

Note that we have calculated the expected rate of return for one year (not four years) to be 12.5%. What if we wanted to estimate the annual rate of return expected over the next five years? In this instance, we might estimate the annual rate of return realized over the previous five years using the geometric mean for that five-year period, for the five years before that, and so forth. Let's say that we estimated 10 such geometric means for nonoverlapping five-year periods going back 50 years. Now, to estimate the expected annual rate of return over the next five years, we would calculate the arithmetic mean of these five-year geometric rates of return.[20]

The idea, again, is that the geometric mean is the appropriate way to measure a historical return. However, if returns are drawn from independent and identically distributed distributions, the best estimator of the mean of that distribution is the arithmetic mean.

rates of return for U.S. stocks and bonds spanning the period 1926–2008. Note that the geometric averages are always lower than arithmetic averages.[20] Although most textbooks use the arithmetic average, practitioners often prefer geometric averages. (For a discussion of geometric versus arithmetic averages, see the Technical Insight box.)

[19] It is obvious that our methodology for estimating five-year returns is limited by the amount of nonoverlapping five-year periods from which we have to choose. The problem gets even worse where we try to estimate the distribution of longer-period returns (say, of 10 or 20 years). In practice, analysts restrict their attention to annual holding period returns because of the convention of quoting returns as annual rates.

[20] This reflects the fact that the geometric mean captures the effects of compound interest, whereas the arithmetic mean does not.

Figure 5 **Historical Stock and Bond Returns: Summary Statistics for 1926–2008**

	Mean		
	Geometric	**Arithmetic**	**Standard Deviation**
Large company stocks	9.6%	11.7%	20.6%
Small company stocks	11.7%	16.4%	3.3%
Long-term corporate bonds	5.9%	6.2%	8.5%
Long-term government bonds	5.7%	6.1%	9.4%
Intermediate-term government bonds	5.4%	5.6%	5.7%
Treasury bills	3.7%	3.8%	3.1%
Inflation	3.0%	3.1%	4.2%

Source: Ibbotson Associates, *SBBI 2009 Yearbook.*

Did You Know?

Do Estimates of the Market Risk Premium Change over Time?

Investment bankers often advise their client firms regarding capital market conditions, including estimates of the market risk premium to use in determining their cost of equity capital. What you may not know is that their estimates of the market risk premium can vary with general economic conditions. For example, from the 1990s up until 2005, the market risk premiums recommended by investment bankers declined from over 7% to as low as 4% in some instances. However, the economic turmoil beginning in 2007 and the uncertainties surrounding the recovery in the years since have led to them using higher market risk premiums in their cost of capital estimates.

Historical data suggest that the equity risk premium for the market portfolio has averaged 6% to 8% a year over the past 75 years. However, there is good reason to believe that, looking forward, the equity risk premium will not be this high. Indeed, current equity risk premium forecasts can be as low as 4%. For the examples in this book, we will use an equity risk premium of 5%, which is commonly used in practice.

CAPM with a size premium The capital asset pricing model (CAPM) is taught in virtually all major business schools throughout the world. Unfortunately, the theory has only weak empirical support. Indeed, academic research has failed to find a significant cross-sectional relation between the beta estimates of stocks and their future rates of return. Research finds that firm characteristics, like market capitalization and book-to-market ratios, provide much better predictions of future returns than do betas.[21] In response to these observations, academics have proposed modifications of the CAPM that incorporate consideration for differences in stock returns related to these firm characteristics.

To illustrate how a size premium might be added to the CAPM, consider the methodology followed by Ibbotson Associates in their 2005 yearbook, which divides firms into four discrete groups based on the total market value of their equity: *Large-cap firms,* those with more than $4.794 billion in market capitalization, receive no size premium. *Mid-cap firms,* with between $4.794 billion and $1.167 billion in equity value,

[21] E. Fama and K. French, "The Cross-Section of Expected Stock Returns," *Journal of Finance* 47 (1992): 427–65.

receive a size premium of 0.91%. *Low-cap firms,* with equity value between $1.167 billion and $331 million, receive a size premium of 1.70%. *Micro-cap firms,* with equity values below $331 million, receive a size premium of 4.01%. Thus, the expected rate of return on equity using the size-adjusted CAPM can be described in Equation 8 below:

$$k_e = k_{rf} + \beta(k_m - k_{rf}) + \begin{cases} \text{Large cap: } 0.0\% \text{ if market cap} > \$4.794 \text{ billion} \\ \text{Mid cap: } 0.91\% \text{ if } \$4.794 \text{ billion} \geq \text{market cap} \geq \$1.167 \text{ billion} \\ \text{Low cap: } 1.70\% \text{ if } \$1.167 \text{ billion} > \text{market cap} > \$.331 \text{ billion} \\ \text{Micro cap: } 4.01\% \text{ if market cap} \leq \$.331 \text{ billion} \end{cases} \quad (8)$$

Factor models A second approach that was introduced in the 1980s is the use of multifactor risk models that capture the risk of investments with multiple betas and factor risk premiums. These risk factors can come from macroeconomic variables, like changes in interest rates, inflation, or GDP, or from the returns of what are known as factor portfolios. Factor portfolios can be formed using purely statistical procedures, like factor analysis or principle components analysis, or by grouping stocks based on their characteristics.

The most widely used factor model is the *Fama-French three-factor model,* which attempts to capture the determinants of equity returns using three risk premiums:[22] the market risk premium of the CAPM, a size risk premium, and a risk premium related to the relative value of the firm when compared to its book value (historical cost-based value). The Fama-French equation for the cost of equity capital includes three factors with their associated risk premiums (thus the use of the term "three-factor model"), found in Equation 9:

$$K_e = K_{rf} + b \times (\text{ERP}) + s \times (\text{SMBP}) + h \times (\text{HMLP}) \quad (9)$$

where k_e is the required rate of return for the common stock of the firm; k_{rf} is the risk-free rate of interest (assumed to be 3%) *b, s,* and *h* are estimated coefficients for the particular firm whose cost of equity is being evaluated; ERP is the equity risk premium, equal to the difference in the expected rate of return from investing in the market portfolio of equities and the long-term risk-free rate of interest (as noted earlier, we use an ERP of 5%); SMBP is the "small minus big risk premium," estimated from historical return differences in large- and small-cap stocks (3.36%); and HMLP is the "high minus low risk premium," estimated as the difference between the historical average annual returns on the high book-to-market (value) and low book-to-market (growth) portfolios (4.4%).[23]

The key to implementing the Fama-French model involves the estimation of the three factor coefficients (i.e., *b, s,* and *h*). To do this, we use the following multiple regression equation of a firm's common stock returns on the historical values of each of the risk premia variables. We illustrate the use of the Fama-French three-factor model for Dell Computer Corporation. Estimates of *b, s,* and *h* are found below, using returns for

[22] E. Fama and K. French, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics* 33 (1993): 3–56.

[23] The equity market risk premium (ERP) used here is our estimate, while the two remaining risk premia represent average risk premia as reported in Ibbotson Associates, *Cost of Capital Yearbook for* 2006, p. 40.

the 48-month period ending with December 31, 2005, and using the regression model in Equation 10.

$$(R_{\text{Dell}} - R_f)_t = \alpha + b(R_m - R_f)_t + s(R_S - R_B)_t + h(R_H - R_L)_t + \epsilon_t \quad (10)$$

where $(R_m - R_f)_t$ is the market's equity risk premium (ERP) for month t, $(R_S - R_B)_t$ is the difference in returns from portfolios of small and large company stock returns for month t (i.e., SMBP), $(R_H - R_L)_t$ is the difference in returns from portfolios of high and low book-to-market stocks for month t (i.e., HMLP), and $\in_t$ is the regression error term. Historical monthly values for the three risk premia variables dating back to 1927 and daily data back to 1963 can be found on Kenneth French's Web site.[24] (See the Technical Insight box on calculating the Fama-French risk premia.)

We estimate Dell's cost of equity by substituting into Equation 10 as follows:

Coefficients	Coefficient Estimates	Risk Premia	Product
b	1.1726	5.00%	5.86%
s	0.1677	3.36%	0.56%
h	−0.7085	4.40%	−3.12%
		Risk premium =	3.31%
		+ Risk-free rate =	3.00%
		Cost of equity	6.31%

The resulting estimate of Dell's cost of equity in this example is 6.31%, which is different from the 9.00% we got using the standard CAPM.

To explore differences between the cost of equity estimates using the Fama-French three-factor model and the standard CAPM, we present the cost of equity using each method for a variety of stocks in Table 2. A quick review of the two estimates of the cost of equity (Fama-French and the traditional CAPM) found in this table indicates that the estimates are very similar for most firms, but there are notable exceptions. For example, the cost of equity from the three-factor model is somewhat lower for Dell and is somewhat higher for General Motors. These differences reflect the fact that the three-factor model is designed to account for the fact that the CAPM has historically overestimated the returns of growth stocks like Dell and underestimated the returns of value stocks like General Motors.

Which model provides a better estimate of the cost of equity? The empirical evidence shows that the three-factor model better explains historical returns than the traditional one-factor CAPM, which is not surprising, given that the model was designed to explain these returns. However, the cost of equity is a forward-looking concept, and we consider historical returns only because they provide some guidance about what we can expect to observe in the future. If we believe that the past is a good indicator of the future, then we should use the three-factor model. However, one might believe that the relatively high returns of value stocks and low returns of growth stocks represent a market inefficiency that is not likely to exist in the future, now that the "value effect" is very well known. In this case, one might prefer the traditional CAPM, which is better grounded in theory.

[24] http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Table 2 **Fama-French Versus CAPM Estimates of the Cost of Equity**

Company Name	Fama-French (FF) Coefficients *b*	*s*	*h*	FF Cost of Equity	CAPM Beta	CAPM Cost of Equity	CAPM Minus FF Cost of Equity
Dell Computers	1.1726	0.1677	−0.7085	6.31%	1.2	9.00%	2.69%
General Motors	0.9189	0.5838	0.9102	13.56%	1.2	9.00%	−4.56%
IBM	1.5145	0.0203	0.3632	12.24%	1.01	8.05%	−4.19%
Merck	0.8035	−1.6398	0.3546	3.07%	0.42	5.10%	2.03%
Pepsico	0.4834	−0.4545	−0.2429	2.82%	0.33	4.65%	1.83%
Pfizer	0.6052	−0.4841	−0.1223	3.86%	0.46	5.30%	1.44%
WalMart	0.6691	−0.2622	−0.0327	5.32%	0.6	6.00%	0.68%

TECHNICAL INSIGHT

Calculating the Fama-French Risk Premia

Annual, monthly, and daily estimates for each of the three sources of risk in the Fama-French three-factor model can be found on Ken French's Web site. Each of the factors is defined as follows:

- $R_m - R_f$, the excess return on the market, is the value-weighted return on all NYSE, AMEX, and NASDAQ stocks (from CRSP) minus the one-month Treasury bill rate (from Ibbotson Associates).
- $R_S - R_B$ is the average return on three small portfolios minus the average return on three big portfolios,

$$R_S - R_B = 1/3\,(\text{Small Value} + \text{Small Neutral} + \text{Small Growth}) - 1/3\,(\text{Big Value} + \text{Big Neutral} + \text{Big Growth})$$

- $R_H - R_L$ is the average return on two value portfolios minus the average return on two growth portfolios,

$$R_H - R_L = 1/2\,(\text{Small Value} + \text{Big Value}) - 1/2\,(\text{Small Growth} + \text{Big Growth})$$

See E. Fama and K. French, "Common Risk Factors in the Returns on Stocks and Bonds," *Journal of Financial Economics* 33 (1993): 3–56, for a complete description of the factor returns.

Limitations of cost of equity estimates based on historical returns The problems that arise with the use of historical returns to estimate the cost of equity relate to the fact that the past returns and, consequently, risk premia are frequently not very reliable predictors of future risk premia. Specifically, there are three fundamental problems associated with using historical returns as the basis for estimating the equity risk premium: (1) Historical returns vary widely over time, resulting in large estimation errors. (2) Recent changes in the tax status of equity returns and the rapid rate of expansion in access to global equity markets are likely to have driven down the equity risk premium. (3) Finally, historical returns over the long term, at least in the United States, may be too high, relative to what we can expect in the future; the U.S. economy did exceptionally well in the last century, and we don't necessarily expect that performance to be repeated.

Historical security returns are highly variable. From Figure 5, we see that the average historical return on a portfolio of large stocks over the period 1926 through 2007 was about 11.7%. Over this period, the average long-term Treasury bond return was 6.1%, implying that the excess return on large stocks was about 5.6%. The standard deviation of these returns was 20.6%, which implies that the standard error of the equity risk premium is approximately 2%.[25] What this means is that the 95% confidence interval of the estimate of the equity risk premium is from 2.5% to 10.5%. Clearly, the breadth of this confidence interval suggests that the most prudent course of action is to interpret historical data as suggestive rather than definitive.

Market conditions are changing. Looking forward, we can say that there are at least three key market factors that have changed over time and that can potentially lead to an equity risk premium that is lower in the future than it was in the past. The first has to do with taxes; the second, with increased participation in the stock market; and the third, with the increased globalization of security markets.

The tax rate on dividend income, interest income, and capital gains fluctuates over time, making equity more or less tax favored in different time periods. When equity is more tax favored, the return premium on equity should be lower relative to the rates on Treasury bonds. In the United States, equity is more tax favored today than at any time in recent history. The Jobs and Growth Tax Relief Reconciliation Act of 2003 reduced the maximum tax rate on dividends to 15%. Previously, dividend income was taxed as ordinary income with rates up to 35%. Thirty years ago, dividends were taxed at 70%.

A second change in market conditions that is likely to affect the equity risk premium in the future relates to the growing number of individuals who participate in the equity market either by direct investments or by indirect investing via their retirement funds. Whereas the equity markets were once dominated by a limited number of wealthy investors, this is no longer the case. An important factor underlying this change was the passage of the Employment Retirement Income Security Act (ERISA) in 1974 and the growth in pension plans that ensued. Today, more than half the equity securities in the United States are held by institutional investors, including mutual funds and pension plans. The broadening participation of more individuals in the public equity markets has led to a sharing of the risk of equity ownership and may have driven down the equity risk premium.

[25] The standard error of the sample mean return is calculated as follows: $\sigma_{\text{Annual Returns}} \div \sqrt{n - 2}$, where the standard deviation in annual returns is 20.2% and n is the number of annual returns in the sample, which here is 78, resulting in an estimate of the standard error of the sample mean of 2.2%.

Still another factor that can lead to a lower equity risk premium in the future relates to effects of improved investor access to global markets. This increased access tends to drive down required rates of return (and consequently the equity risk premium) due to the increased opportunities for diversification and greater participation by more investors who share the risks associated with equity investments.[26]

Historical returns exhibit survivor bias. The final problem that we encounter when using historical returns to estimate future equity risk premia relates to the fact that the historical market returns that are generally available reflect the performance of stock markets that have done relatively well. That is, only the winners in the financial markets, such as the United States and the United Kingdom, survive to be analyzed and incorporated into the record of historical returns. Indeed, there are very few markets that have an uninterrupted period of surviving for over 75 years. History teaches us that we cannot confidently forecast that the next 75 years will be equally successful.

Method 2—Discounted Cash Flow or Imputed Rate of Return Approaches

Up to this point, we have estimated expected rates of return using the past return history as our guide. However, as the above discussion indicates, past returns may be an overly optimistic indicator of expected future returns. In this section, we will describe a method of estimating the return on the market that uses *forward-looking* rather than *historical* estimates of *equity risk premia.*

Our forward-looking approach starts with the familiar discounted cash flow model. However, instead of using the DCF model to determine the value of an investment, the method takes observed values and estimated cash flows and uses the DCF model to estimate the internal rate of return, or the *implied* cost of equity capital. As we discuss below, analysts apply this methodology in two related ways. Some use the method to come up with an estimate of the market return premium that can be used to generate a CAPM-based estimate of the cost of equity. Others use the method to directly estimate a forward-looking cost of equity for individual firms.

Single-stage DCF growth model We discuss two variants of the discounted cash flow model. The first is the Gordon growth model, which assumes that the firm's dividends grow at a constant rate forever (i.e., that there is a single growth rate or single stage of growth). The resulting single-stage discounted cash flow model of equity value can easily be deduced from the general discounted cash flow model found in Equation 11a below:

$$\text{Stock Price}_0 = \sum_{t=1}^{\infty} \frac{\text{Div}_{\text{Year}\,t}}{(1 + k_e)^t} \tag{11a}$$

where

Stock Price_0 = the current stock price of the firm's shares,
$\text{Div}_{\text{Year}\,t}$ = expected dividend for year t, and
k_e = the cost of equity capital.

[26] For example, Rene Stulz suggests that the increased investment opportunity set would reduce the expected risk premium to about two-thirds of the average since 1926 (Rene Stulz, "Globalization of Capital Markets and the Cost of Capital: The Case of Nestle," *Journal of Applied Corporate Finance* 8, no. 3 (Fall 1995).

When the firm's dividends are expected to grow at a constant rate, g, forever and this rate is less than the firm's cost of equity capital, the above valuation expression can be reduced to the following:

$$\text{Stock Price}_0 = \frac{\text{Div}_{\text{Year}\,0}(1+g)}{k_e - g} = \frac{\text{Div}_{\text{Year}\,1}}{k_e - g} \tag{11b}$$

Consequently, the cost of equity can be found by solving for k_e in Equation 11c:

$$k_e = \frac{\text{Div}_{\text{Year}\,1}}{\text{Stock Price}_0} + g \tag{11c}$$

Analysts typically solve for k_e by observing the most recent dividend paid over the last 12 months, $\text{Div}_{\text{Year 0}}$, along with the most recent Stock Price_0 for the company's shares, and using analysts' earnings growth rate estimates to estimate the growth rate in the firm's dividends. Depending on how the cost of equity estimate is to be used, firms will either use dividend growth estimates from stock analysts who cover the firm or will use their own internal estimates.

To illustrate, consider the case of Duke Energy Corporation (DUK). Duke is involved in a number of businesses, including natural gas transmission and electric power production. In 2008, the company paid a dividend of \$0.96 per share, and on November 5, 2009, the firm's stock closed trading at a price of \$15.98. The analysts' expected rate of growth in earnings for 2010 through 2014 is 3.2% per annum.[27] Because Equation 11c assumes that the firm's dividends grow forever at a constant rate, we use the five-year estimate to estimate Duke's cost of equity capital as follows:

$$k_e = \frac{\text{Div}_{\text{Year 0}}(1+g)}{P_E} + g = \frac{\$0.96(1+.032)}{\$15.98} + .032 = .093997 \text{ or } 9.4\%$$

Aggregating the forward-looking equity risk premia The method we have just used to impute the equity required rate of return for one firm can also be used to estimate the market's equity risk premium. The method requires that the equity risk premia for a broad sample of individual firms be estimated and then aggregated across the sample. A number of recent studies have performed this analysis, and the principal feature of the empirical evidence regarding forward-looking equity risk premia is that they are much lower than the estimates based on historical data.[28] In general, most financial analysts who estimate forward-looking equity risk premia find them to be in the 3% to 5% range.

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27 Based on the Yahoo Finance five-year earnings growth rate estimate (http://finance.yahoo.com/q/ae?s=DUK).

28 Specifically, forward equity risk premia estimated in recent years for the U.S. equity market have been 3% or less. Moreover, some researchers have argued that the historical estimates of an 8% equity risk premium (using the Treasury bills to proxy for the risk-free rate) and 5% (where long-term U.S. Treasury bonds are used to proxy the risk-free rate) have never been realistic except at market bottoms or at times of crisis (i.e., war).

BEHAVIORAL INSIGHT

Managerial Optimism and the Cost of Capital

The tendency for managers to be overly optimistic about their own firm's prospects is a well-documented behavioral bias. However, this bias may not be a serious problem when it comes to the estimation of the firm's cost of capital if a discounted cash flow model such as the Gordon growth model is used. This approach to evaluating a firm's cost of equity has a potentially important side benefit when the cost of equity is estimated by a member of the firm's management. Very simply, overoptimism on the part of a member of the management team in estimating the anticipated rate of growth in the firm's earnings and dividends will lead to an upward bias in the cost of equity and, consequently, the cost of capital estimate.

To illustrate, let's consider the estimate of Duke Energy Corporation's (DUK) cost of equity from the above example. If Duke's internal analysts are making their own estimates about future growth in the firm's earnings and dividends, they may feel that 5% is a reasonable estimate, whereas market analysts used 3.2%. The effect of Duke's optimistic estimate of the future rate of growth in earnings and dividends is to increase the cost of equity from 9.4% to 11.3%. This higher discount rate will tend to counter the effects of excessive optimism that these same analysts may exhibit when estimating the future cash flows of Duke's investment opportunities.

Three-stage growth model The second variant of the discounted cash flow model differs from the first only in that it provides for *three* different growth rates, corresponding to three stages of a company's growth. Specifically, this model provides for different dividend growth rates for Years 1 through 5, 6 through 10, and 11 and beyond. The corresponding three-stage growth model can be written as follows:

$$\begin{aligned}\text{Stock Price}_{\text{Year}_0} = {} & \sum_{t=1}^{5} \frac{\text{Div}_{\text{Year }0}(1+g_{\text{Years 1 to 5}})^t}{(1+k_e)^t} \\ & + \sum_{t=6}^{10} \frac{\text{Div}_{\text{Year }0}(1+g_{\text{Years 1 to 5}})^5(1+g_{\text{Years 6 to 10}})^{t-5}}{(1+k_e)^t} \\ & + \left(\frac{\text{Div}_{\text{Year }0}(1+g_{\text{Years 1 to 5}})^5(1+g_{\text{Years 6 to 10}})^5(1+g_{\text{Years 11 and beyond}})}{k_e - g_{\text{Years 11 and beyond}}}\right) \\ & \times \frac{1}{(1+k_e)^{10}} \qquad (12)\end{aligned}$$

Given estimates of the three growth rates ($g_{\text{Years 1 to 5}}$, $g_{\text{Years 6 to 10}}$, and $g_{\text{Years 11 and beyond}}$), the current period dividend ($\text{Div}_{\text{Year 0}}$), and the current stock price ($\text{Stock Price}_{\text{Year 0}}$), we can solve for the cost of equity (k_e).

The advantage of the three-stage model is its flexibility to incorporate different growth rates over the life cycle of the firm. The corresponding disadvantage, of course, is that it requires that these growth rates be estimated. Ibbotson Associates uses the I/B/E/S expected rate of growth in earnings to estimate the first growth rate, and the average rate of historical growth in earnings for the firm's industry to estimate the second. For the third

Practitioner Insight

Market-Implied (ex ante) Market Risk Premia—An Interview with Justin Pettit*

Wall Street analysts estimate the equity risk premium (hereafter ERP) in one of two ways. The ERP is typically estimated using the average spread of historical equity returns over a U.S. Treasury yield. Because this method uses only historical data, we can think of it as the ex post method. The ERP is also estimated using professional analysts' expectations of future performance and the Gordon growth model (see Equation 11b), which is the ex ante method.

The Gordon dividend discount model is a single-stage growth model (see Equation 11b) that can be rewritten for purposes of estimating the ERP. If we apply the Gordon model to the market for all common equity and then solve for the required rate of return on equity and subtract the risk-free rate of interest, we estimate the ERP as follows:

$$\text{ERP} = \left(\frac{\text{Div}_{\text{Year }0}(1 + g)}{\text{Market Cap}} + g \right) - k_f$$

where $\text{Div}_{\text{Year }0}$ is the most recent year's annual dividend payment for the market as a whole, Market Cap is the total market value of all equities, and g is the estimated long-term dividend growth rate for all stocks in the economy. Because the constant growth rate assumption of the single-stage Gordon model is problematic for a single company, analysts feel that it is more useful for a broad market.[29]

Analysts typically estimate the growth rate as the product of the average return on equity (ROE) and reinvestment rate for all equities where the reinvestment rate is simply 1 minus the fraction of firm earnings distributed to shareholders through dividends and

growth rate, it uses a rate that reflects the long-term rate of growth in GDP and long-term inflation forecast; for 2004, this growth rate was 3%.[30]

To illustrate the methodology, consider Rushmore Electronics, which is a small but fast-growing company headquartered in Portland, Oregon. The firm went public two years ago, and the firm's stock currently sells for $24.00 per share, with cash dividends per share of $2.20. Over the past 10 years, Rushmore's earnings grew at a compound annual rate of 10% per year. Analysts project that the firm's earnings will grow at a rate

[29] The ex ante model described here follows a "top-down" approach because it focuses on the aggregate return to the market for equities as a whole. An alternative method that we discuss in the text begins by estimating the ERP for individual stocks and then aggregating them to form an estimate of the ERP for all equities. This is the "bottom-up" approach.

[30] One further caveat needs to be mentioned with regard to Ibbotson's three-stage growth model. Because many firms do not pay dividends, Ibbotson Associates replaces dividends with cash flow in the first two terms and uses earnings per share before extraordinary items in the third term. In the latter case, the earnings substitution for cash flow reflects the assumption that, over time, depreciation and capital expenditures will be roughly equal, such that cash flow is equal to earnings.

share repurchases. This is an ex ante approach to estimating future growth rates because it uses current market information to impute the market's expected MRP. For example, a 10% ROE and 65% reinvestment rate implies a 6.5% growth rate. Note that we are using the "distributed" rather than the "dividend" yield, for, increasingly, firms pay out cash via share repurchases rather than dividends. Regardless of the mechanism, these funds are not reinvested.

The following table shows a range of potential market risk premia as a function of assumed perpetual growth rates and distributed yields (i.e., ratios of cash distributions through dividends and share repurchases divided by the current market capitalization):

Market Risk Premium		**Nominal Perpetual Growth Rate**				
		4%	5%	6%	7%	8%
Distributed Yield	1%	0.3%	1.3%	2.3%	3.3%	4.3%
	2%	1.3%	2.3%	3.3%	4.3%	5.3%
	3%	2.3%	3.3%	4.3%	5.3%	6.3%
	4%	3.3%	4.3%	5.3%	6.3%	7.3%

Analysts typically arrive at an estimate of 4% to 5% for the MRP under growth-rate assumptions of 5% to 7% and distributed yields of 3% to 4%. The 5% to 7% consensus estimates of long-term sustainable (nominal) growth rates are consistent with expected inflation of 2% to 3% and real GDP growth of 3% to 4%.

*Justin Pettit is a partner, Booz Allen Hamilton, New York, New York.

of 14% per year over the next five years. Moreover, the economy is expected to grow at a rate of 6.5%. Substituting into Equation 12, we get the following equation, in which the only unknown is the cost of equity, k_e:

$$\$24.00 = \sum_{t=1}^{5} \frac{\$2.20(1 + .14)^t}{(1 + k_e)^t} + \sum_{t=6}^{10} \frac{\$2.20(1 + .14)^5(1 + .10)^{t-5}}{(1 + k_e)^t} + \left(\frac{\$2.20(1 + .14)^5(1 + .10)^5(1 + .065)}{k_e - .065} \right) \frac{1}{(1 + k_e)^{10}}$$

Solving for k_e produces an estimate of Rushmore's cost of equity of 20.38%.[31]

In this example, we have followed the Ibbotson Associates approach by making the first two growth periods five years long and by using their prescription for the estimation of stages two and three growth rates. If the analyst has good reason to believe that either

[31] This problem is easily solved in a spreadsheet using built-in functions (for example, in Excel, the function is "Goal Seek").

the length of the growth periods or the specific growth rates should be different, these estimates should be used.

Calculating the WACC (Putting It All Together)—Step 3

The final step in the estimation of a firm's WACC involves calculating a weighted average of the estimated costs of the firm's outstanding securities. To illustrate the three-step process used to estimate WACC, consider the case of Champion Energy Corporation, which was founded in 1987 and is based in Houston, Texas. The company provides midstream energy services, including natural gas gathering, intrastate transmission, and processing in the southwestern Louisiana and Texas Gulf Coast regions. The company operates approximately 5,500 miles of natural gas gathering and transmission pipelines and seven natural gas processing plants.

Evaluating Champion's Capital Structure

A condensed version of Champion's liabilities and owner's equity is found in the first two columns of Table 3.

Champion wants to reevaluate its cost of capital in light of its plans to make a significant acquisition to expand its operations in January and needs an additional $1.25 billion. Based on discussions with the firm's investment banker, Champion's

Table 3 Liabilities and Owner's Equity for Champion Energy Corporation

Liabilities and Owner's Capital ($000)	December 31, 2009 Balance Sheet (Book Values)	Invested Capital (Market Values)
Current liabilities		
Accounts payable	$ 150,250.00	
Notes payable	—	$ —
Other current liabilities	37,250.00	
Total current liabilities	$ 187,500.00	$ —
Long-term debt (8% interest paid semiannually, due in 2015)	750,000.00	2,000,000.00
Total liabilities	$ 937,500.00	2,000,000.00
Owner's capital		
Common stock ($1 par value per share)	$ 400,000.00	
Paid-in-capital	1,250,000.00	
Accumulated earnings	2,855,000.00	
Total owner's capital	$4,505,000.00	8,000,000.00
Total liabilities and owner's capital	$5,442,500.00	$10,000,000.00

management has chosen to increase the firm's debt from the current level of $0.75 billion up to $2 billion. Given current market conditions, they can raise the additional debt at 8.25%. Increasing the firm's use of borrowed funds is appealing to Champion's management because the interest expense offsets taxable income at the 25% tax rate faced by the firm.

Champion's CFO called in his chief financial analyst and asked that she make an initial estimate of the firm's WACC, assuming the firm went through with the $1.25 billion debt offering and assuming that the nature of the firm's operations and business risk are not affected by the use of the net proceeds. The analyst began her analysis by evaluating the firm's capital structure under the assumption that the new debt offering was consummated. Because the effect of the offering on the value of the firm's equity was uncertain, she thought, as a first approximation, that she would just assume that the total market value of the firm's shares would be unchanged. Based on the current market price of $20.00 per share, she calculated the market capitalization of the firm's equity to be $8 billion.

The third column of Table 3 contains the results of the analyst's investigation into Champion's proposed capital structure following the new debt issue. Champion's total invested capital is equal to $10 billion, which includes interest-bearing debt of $2 billion and the firm's equity capitalization of $8 billion ($20.00 per share multiplied by 400 million shares). Consequently, the capital structure weights are 20% debt and 80% equity.

Estimating Champion's Costs of Debt and Equity Capital

Based on current yields to maturity for Champion's new debt offering, we estimate the before-tax cost of debt financing to be 8.25%. Because Champion enjoys an investment-grade bond rating, we can use the yield to maturity as a reasonable approximation to the cost of new debt financing. Adjusting the 8.25% yield for the firm's 25% tax rate produces an after-tax cost of debt of 6.19% = 8.25%(1 − .25).

To calculate the cost of equity, three estimates were used: the CAPM, the three-factor Fama-French, and the three-stage DCF models.[32] The resulting estimates are found below:

Cost of Equity Model	**Estimated Cost of Equity**
CAPM	8.37%
Three-factor Fama-French	10.02%
Discounted cash flow (three-stage)	11.60%

The average of the three estimates of the cost of equity is 10.0%, and the analyst decided to use this as her initial estimate of the cost of equity capital to calculate the firm's WACC.

[32] These cost of equity estimates represent the large industry composite for SIC 4924 (Natural Gas Distribution), as estimated by Ibbotson and Associates, *Cost of Capital 2008 Yearbook* (data through March 2008).

Calculating Champion's WACC

The WACC for Champion's proposed capital structure is calculated as follows:

Source of Capital	Capital Structure Weight (Proportion)	After-Tax Cost	Weighted After-Tax Cost
Debt	20%	6.19%	0.01238
Equity	80%	10.00%	0.07997
		WACC =	9.23%

Therefore, we estimate Champion's WACC to be 9.23%, based on its planned use of debt financing and operating plans.

Taking a Stand on the Issues—Estimating the Firm's Cost of Capital

Our discussion of the estimation of the cost of capital began by enumerating three basic issues that had to be addressed. In the process of reviewing all the ins and outs of the estimation process, it is apparent that analysts must make a lot of decisions that can have a material effect on the cost of capital estimate. Here we briefly recap each of the issues that must be addressed and summarize the procedures that we feel represent the best thinking on each issue.

Figure 6 compiles a listing of each of the basic issues we have addressed in the estimation of the cost of capital to this point. The salient points are the following:

- Use market value weights to define a firm's capital structure and omit non-interest-bearing debt from the calculations. If the firm plans to change its current usage of debt and equity, then the target weights should replace current weights.
- If the cost of capital is to be used to discount distant future cash flows, then use the yield on a long-term bond in the estimation of the market risk premium, as well as in calculating the excess market returns that are used in the estimation of beta.
- When the firm issues investment-grade debt, use the yield to maturity (estimated using current market prices and promised interest and principal payments) to estimate the cost of debt. However, when the firm's debt is speculative-grade, the yield to maturity on the firm's debt (which represents the promised, not expected, yield on the debt) will overestimate the cost of debt financing.
- Use multiple methods for estimating the cost of equity capital. This is the single most difficult estimate the firm will make. Using both asset pricing models and discounted cash flow models provides independent estimates of the cost of equity.
- Because the cost of capital is used to discount cash flows to be received in the future, the focus of our analysis should be forward-looking. This does not mean that we ignore historical data. However, it does mean that historical data are useful only if they help us better understand the future. Indeed, the equity risk premium for the market is typically estimated as an average of past returns, which implies a market risk premium of 6% to 8%. However, a number of recent studies have utilized the expectations of investment professionals to infer an equity risk premium and found it to be much lower. The estimates for the forward-looking market risk premium are in the 3%-to-4% range.

Figure 6 **Issues and Recommendations on the Estimation of a Firm's Cost of Capital**

Topic	Issues	Best Practices
(1) Defining the firm's capital structure	What liabilities should be included in defining the firm's capital structure?	Include only those liabilities that have an explicit interest cost associated with them. Specifically, exclude non-interest-bearing liabilities such as accounts payable.
	How should the various sources of capital in the capital structure be weighted?	Weights should reflect the current importance of sources of financing, which, in turn, is reflected in current market values. However, because corporate bonds do not trade often, if at all, and a large component of corporate debt is private debt (i.e., bank loans), which does not have an observable market value, book values are often used.
(2) Choosing the appropriate risk-free rate of interest	What maturity is appropriate for the risk-free government security?	U.S. government debt is the best representation of a risk-free security. However, asset pricing theory provides little guidance as to whether a short-term, intermediate-term, or long-term U.S. Treasury security should be used. Standard practice now is to use a long-term government bond.
(3) Estimating the cost of debt financing	How do you estimate the expected rate of return on investment-grade debt?	Current yields to maturity serve as a reasonable proxy for the expected cost of debt.
	How do you estimate the expected rate of return on speculative or below-investment-grade debt?	Adjustments for the risk of default and recovery rates become important when the prospect of default is significant. The adjustments lead to an estimate of the expected rate of return to the firm's creditors that is based on expected cash flows rather than promised cash flows.
(4) Estimating the cost of equity financing	What model should be used to estimate the cost of equity financing?	Two classes of models are widely used in the estimation of the cost of equity financing: One is based on asset pricing theory, and the other is based on discounted cash flow. Given the inherent difficulties involved in estimating the cost of equity capital, it is wise to use both types of models in an effort to define the potential range of equity capital costs.
	Should historical or forward-looking data be the basis for estimating the cost of common equity?	Historical data are useful only in estimating the cost of capital to the extent that these data are informative about future returns. Thus, in general, forward-looking information is more consistent with the objective of the analysis than are historical data.
	How large is the equity risk premium?	Historical data suggest that the equity risk premium for the market portfolio has averaged 6% to 8% a year over the past 75 years. However, there is good reason to believe that this estimate is far too high. In fact, the equity risk premium according to recent estimates lies in the range of 3% to 4%. We recommend a 5% equity risk premium for the market.

4 Summary

A firm's weighted average cost of capital (WACC) provides the rate that is used to discount a firm's future cash flows and determine how it is likely to be valued in the financial marketplace. The estimation of a firm's WACC involves three fundamental activities: evaluating the composition of the firm's capital structure, estimating the opportunity cost of each source of capital, and calculating a weighted average of the after-tax cost of each source of capital.

The WACC is used extensively by firms throughout the world. Firms use their own WACCs to calculate whether they are under- or overvalued. Hence, a firm's WACC will influence how other firms respond to acquisition offers, and because firms are reluctant to issue undervalued securities, the WACC calculation can also influence financing choices. Finally, firms often use their WACC to evaluate the extent to which their managers are able to generate returns that exceed their firms' cost of capital.

The main focus in this book is on how firms evaluate investment opportunities. In this regard, the WACC plays a key role. When firms evaluate opportunities to acquire other firms, they calculate the WACC of the acquisition candidate. When firms evaluate an investment project, they need a discount rate that we will refer to as the project WACC.

Problems

1 CALCULATING A FIRM'S WACC Nestle Enterprises is estimating its cost of capital for the first time and has made the following estimates: The firm's debt carries an AAA rating, which is currently yielding 6%; the firm pays taxes at a rate of 30%; the cost of equity is estimated to be 14%; and the firm's debt is equal to 20% of its enterprise value. What is Nestle's WACC?

2 CALCULATING THE COST OF EQUITY Caliber's Burgers and Fries is a rapidly expanding chain of fast-food restaurants, and the firm's management wants to estimate the cost of equity for the firm. As a first approximation, the firm plans to use the beta coefficient for McDonald's Corp. (MCD), which equals .56 as a proxy for its beta. In addition, Caliber's financial analyst looked up the current yield on 10-year U.S. Treasury bonds and found that it was 4.2%. The final piece of information needed to estimate the cost of equity using the capital asset pricing model is the market risk premium, which is estimated to be 5%.

a. Using this information, estimate the cost of equity for McDonald's.
b. McDonald's Corp. has an enterprise value of about \$80 billion and a debt of \$15 billion. If Caliber's has no debt financing, what is your estimate of the firm's beta coefficient? You can assume that McDonald's debt has a beta of .20.

3 ESTIMATING THE UNLEVERED BETA The CFO of Sterling Chemical is interested in evaluating the cost of equity capital for his firm. However, Sterling uses very little debt in its capital structure (the firm's debt-to-equity capitalization ratio is only 20%), while

larger chemical firms use substantially higher amounts of debt. The following table shows the levered equity betas, debt-to-equity betas, and debt betas for three of the largest chemical firms.

Company Name	Levered Equity Betas	Debt/Equity Capitalization	Assumed Debt Betas
Eastman Chemical Co. (EMN)	2.0200	42.69%	0.30
Celanese Corp. (CE)	2.6300	82.93%	0.30
Dow Chemical Company (DOW)	2.5000	29.07%	0.30

a. Use the information given above to estimate the unlevered equity betas of each of these companies.
b. If Sterling's debt-to-equity capitalization ratio is .20 and its debt beta is also .30, what is your estimate of the firm's levered equity beta?

4 THREE-STEP PROCESS FOR ESTIMATING A FIRM'S WACC Compano Inc. was founded in 1986 in Baytown, Texas. The firm provides oil-field services to the Texas Gulf Coast region, including the leasing of drilling barges. Its balance sheet for year-end 2006 describes a firm with $830,541,000 in assets (book values) and invested capital of more than $1.334 billion (based on market values):

	December 31, 2006	
Liabilities and Owners' Capital	Balance Sheet (Book Values)	Invested Capital (Market Values)
Current liabilities		
Accounts payable	$ 8,250,000	
Notes payable	—	—
Other current liabilities	7,266,000	
Total current liabilities	$ 15,516,000	$ —
Long-term debt (8.5% interest paid semiannually, due in 2015)	$ 420,000,000	$ 434,091,171
Total liabilities	$ 435,516,000	$ 434,091,171
Owners' capital		
Common stock ($1 par value per share)	$ 40,000,000	
Paid-in-capital	100,025,000	
Accumulated earnings	255,000,000	
Total owners' capital	$395,025,000	$ 900,000,000
Total liabilities and owners' capital	$830,541,000	$1,334,091,171

Compano's executive management team is concerned that its new investments be required to meet an appropriate cost of capital hurdle before capital is committed. Consequently, the firm's CFO has initiated a cost of capital study by one of his senior financial analysts, Jim Tipolli. Jim's first action was to contact the firm's investment banker to get input on current capital costs.

Jim learned that although the firm's current debt capital required an 8.5% coupon rate of interest (with annual interest payments and no principal repayments until 2015), the current yield on similar debt had declined to 8% if the firm were to raise debt funds today. When he asked about the beta for Compano's debt, Jim was told that it was standard practice to assume a beta of .30 for the corporate debt of firms such as Compano.

a. What are Compano's total invested capital and capital structure weights for debt and equity?

b. Based on Compano's corporate income tax rate of 40%, the firm's current capital structure, and an unlevered beta estimate of .90, what is Compano's levered equity beta?

c. Assuming a long-term U.S. Treasury bond yield of 5.42% and an estimated market risk premium of 5%, what should Jim's estimate of Compano's cost of equity be if he uses the CAPM?

d. What is your estimate of Compano's WACC?

5 CALCULATING THE EXPECTED YTM International Tile Importers Inc. is a rapidly growing firm that imports and markets floor tiles from around the world that are used in the construction of custom homes and commercial buildings. The firm has grown so fast that its management is considering the issuance of a five-year interest-only note. The notes would have a principal amount of $1,000 and pay 12% interest each year, with the principal amount due at the end of Year 5. The firm's investment banker has agreed to help the firm place the notes and has estimated that they can be sold for $800 each under today's market conditions.

a. What is the promised yield to maturity based on the terms suggested by the investment banker?

b. (*Hint:* Refer to the Appendix for this analysis.) The firm's management looked with dismay at the yield to maturity estimated above, for it was much higher than the 12% coupon rate, which is much higher than current yields on investment-grade debt. The investment banker explained that for a small firm such as International Tile, the bond rating would probably be in the middle of the speculative grades, which requires a much higher yield to attract investors. The banker even suggested that the firm recalculate the expected yield to maturity on the debt under the following assumptions: The risk of default in Years 1 through 5 is 5% per year, and the recovery rate in the event of default is only 50%. What is the expected yield to maturity under these conditions?

6 CALCULATING THE PROMISED YTM In 2005, the Eastman Kodak Corporation (EK) had a straight bond issue outstanding that was due in eight years. The bonds are currently selling for 108.126 percent, or $1,081.26, per bond and pay a semiannual interest payment

based on a 7.25% (annual) coupon rate of interest. Assuming that the bonds remain outstanding until maturity and the company makes all promised interest and principal payments in a timely basis, what is the yield to maturity to the bondholders?

7 CALCULATING THE PROMISED YTM Evaluate the promised yield to maturity for the bonds issued by Ford (F) and General Motors (GM) as of February 2005. (You may assume that interest is paid semiannually. Also, round the number of compounding periods to the nearest six months.)

Ford Motor Co.	
Coupon:	6.3750%
Maturity:	02/01/2029
Rating:	Baa1/BBB–
Price:	92.7840
General Motors Corp.	
Coupon:	8.375%
Maturity:	07/15/2033
Rating:	Baa2/BBB–
Price:	106.1250

Noting that both these bond issues have similar ratings, comment on the use of the yield to maturity you have just calculated as an estimate of the cost of debt financing to the two firms.

8 CALCULATING THE COST OF EQUITY Smaltz Enterprises is currently involved in its annual review of the firm's cost of capital. Historically, the firm has relied on the CAPM to estimate its cost of equity capital. The firm estimates that its equity beta is 1.25, and the current yield on long-term U.S. Treasury bonds is 4.28%. The firm's CFO is currently in a debate with one of the firm's advisors at its investment bank about the level of the equity risk premium. Historically, Smaltz has used 7% to approximate the equity risk premium. However, the investment banker argues that this premium has shrunk dramatically in recent years and is more likely to be in the 3%-to-4% range.

- **a.** Estimate Smaltz's cost of equity capital using a market risk premium of 3.5%.
- **b.** Smaltz's capital structure is comprised of 75% equity (based on current market prices) and 25% debt on which the firm pays a yield of 5.125% before taxes at 25%. What is the firm's WACC under each of the two assumptions about the market risk premium?

9 THREE-STEP PROCESS FOR ESTIMATING A FIRM'S WACC Harriston Electronics builds circuit boards for a variety of applications in industrial equipment. The firm was founded in 1986 by two electrical engineers who left their jobs with the General Electric (GE)

Corporation. Its balance sheet for year-end 2009 describes a firm with $1,184,841,000 in assets (book values) and invested capital of approximately $2.2 billion (based on market values):

	December 31, 2009	
Liabilities and Owners' Capital	**Balance Sheet (Book Values)**	**Invested Capital (Market Values)**
Current Liabilities		
Accounts payable	$ 17,550,000	
Notes payable	20,000,000	20,000,000
Other current liabilities	22,266,000	
Total current liabilities	$ 59,816,000	$ 20,000,000
Long-term debt (7.5% interest paid semiannually, due in 2015)	$ 650,000,000	$ 624,385,826
Total liabilities	$ 709,816,000	$ 644,385,826
Owners' capital		
Common stock ($1 par value per share)	$ 20,000,000	
Paid-in-capital	200,025,000	
Accumulated earnings	255,000,000	
Total owners' capital	$ 475,025,000	$1,560,000,000
Total liabilities and owners' capital	$1,184,841,000	$2,204,385,826

Harriston's CFO, Margaret L. Hines, is concerned that its new investments be required to meet an appropriate cost of capital hurdle before capital is committed. Consequently, she initiated a cost of capital study by one of her senior financial analysts, Jack Frist. Shortly after receiving the assignment, Jack called the firm's investment banker to get input on current capital costs.

Jack learned that although the firm's current debt capital required a 7.5% coupon rate of interest (with annual interest payments and no principal repayments until 2015), the current yield to maturity on similar debt had risen to 8.5%, such that the current market value of the firm's outstanding bonds had fallen to $624,385,826. Moreover, because the firm's short-term notes were issued within the last 30 days, the 9% contract rate on the notes was the same as the current cost of credit for such notes.

a. What are Harriston's total invested capital and capital structure weights for debt and equity? (*Hint:* The firm has some short-term debt [notes payable] that is also interest bearing.)

b. Assuming a long-term U.S. Treasury bond yield of 5.42% and an estimated market risk premium of 5%, what is Harriston's cost of equity based on the CAPM if the firm's levered equity beta is 1.2?

c. What is your estimate of Harriston's WACC? The firm's tax rate is 35%.

10 UNLEVERING AND RELEVERING EQUITY BETAS In 2006, the major airline carriers, with the principal exception of Southwest Airlines (LUV), continued to be in dire financial condition following the attack on the World Trade Center in 2001.

a. Given the following data for Southwest Airlines and three other airlines (for November 28, 2006), estimate the unlevered equity beta for Southwest Airlines using the procedure described in Table 1. You may assume a 38% tax rate in your calculations.

Company Name	Levered Equity Beta	Debt/Equity Capitalization	Assumed Debt Beta
American Airlines (AMR)	3.2400	205.16%	0.30
Delta Airlines (DALR.PK)	4.0500	5663.67%	0.40
Jet Blue (JBLU)	(0.1100)	106.22%	0.30
Southwest Airlines (LUV)	(0.0100)	14.93%	0.20

b. Based on your estimate of Southwest Airlines's unlevered equity beta, relever the beta to get an estimate of the firm's levered beta.

c. The airline industry was obviously in a very unique position at the end of 2006. Are there any special concerns that you have regarding the estimation of the cost of equity for Southwest Airlines using the procedure described here?

11 ESTIMATING THE COST OF EQUITY USING THE FAMA-FRENCH METHOD Telecom services is an industry under rapid transformation as telephone, Internet, and television services are being brought together under a common technology. In the fall of 2006, the telecom analyst for HML Capital, a private investment company, was trying to evaluate the cost of equity for two giants in the telecom industry: SBC Communications (AT&T) and Verizon Communications. Specifically, he wanted to look at two alternative methods for making the estimate: the CAPM and Fama-French three-factor model. The CAPM utilizes only one risk premium for the market as a whole, whereas the Fama-French model uses three (one for each of three factors). The factors are (1) a market risk premium, (2) a risk premium related to firm size, and (3) a market to book risk premium. Data for the risk premium sensitivities (*b, s,* and *h*) as well as the beta coefficient for the CAPM are found below:

	Fama-French (FF) Coefficients			
Company Name	***b***	***s***	***h***	**CAPM Beta**
SBC Communications (AT&T)	1.0603	−1.4998	1.0776	0.62
Verizon Communications	1.1113	−0.9541	0.639	0.79

The risk premia for each of the risk factors are as follows:

Coefficients	Risk Premia
Market risk premium	5.00%
Size risk premium	3.36%
Market to book risk premium	4.40%

a. If the risk-free rate of interest is 3%, what is the estimated cost of equity for the two firms using the CAPM?

b. What is the estimated cost of equity capital for the two firms using the Fama-French three-factor model? Interpret the meaning of the signs (+/−) attached to each of the factors (i.e., *b, s,* and *h*).

12 CALCULATING THE COST OF CONVERTIBLE DEBT (*Hint:* This exercise is based on the Appendix.) The Eastman Kodak Corporation had an issue of convertible bonds outstanding in the spring of 2005 that had a 6.75% coupon rate of interest and sold for $1,277.20 per bond. The interest payments were made semiannually. If the cost of straight debt was equal to the yield to maturity on the bonds described in Problem 3 and the cost of new equity raised through the sale of the conversion option was 13.5%, what was the estimated cost of convertible bonds to the company?

13 PROMISED VERSUS EXPECTED YTM (*Hint:* This exercise is based on the Appendix.) In May 2005, the credit rating agencies downgraded the debt of the General Motors Corporation (GM) to junk status.

a. Discuss the effect of this drop in credit status on the company's WACC.
b. If you have been using the yield to maturity for the firm's bonds as your estimate of the cost of debt capital and you use the same estimation procedure following the downgrade, has your estimate become more or less reliable? Explain.

APPENDIX

Extensions and Refinements of WACC Estimation

Our analysis of WACC in the chapter dealt with the basic issues encountered in a simple setting. In this section, we address some problems that arise in the analysis of a firm's WACC that are often overlooked in introductory discussions. Specifically, we discuss: (1) the estimation of the cost of below-investment-grade debt, where the promised yield to maturity is often much higher than the expected cost of the debt; (2) estimation of the cost of convertible debt, which is a hybrid form of financing that contains both debt and equity elements; and (3) the analysis of *off-balance-sheet* sources of financing, including special-purpose entities.

Estimating the Expected Cost of Below-Investment-Grade Debt Financing

We noted earlier in the chapter that the promised yield to maturity (calculated using the promised principal and interest payments) overstates the cost of debt to the company if the debt is below investment grade. To see why this is so, consider how the stated yield to maturity of a bond is calculated. The yield to maturity, as traditionally defined, is simply the internal rate of return earned by purchasing a bond for its current market price, assuming that you receive all of the bond's promised interest and principal payments over the life of the bond. Because this calculation uses *promised* interest and principal payments and not *expected* cash receipts (which allow for the prospect of default and the recovery of less than 100% of the bond's face value), this computation produces an estimate of the *promised yield to maturity* on the debt,[33] not the *expected yield to maturity.*

We calculate the promised yield to maturity by solving for the discount rate, Y_{Promised}, which satisfies the following discounted cash flow equation:

$$\begin{aligned}\begin{matrix}\text{Bond Price}\\ \text{Today}\end{matrix} = \$1{,}000 &= \frac{\text{Promised Principal and Interest in One Year}}{(1 + Y_{\text{Promised}})^1}\\ &= \frac{\$1{,}100}{(1 + Y_{\text{Promised}})^1}\end{aligned}$$

[33] We have used the term "hoped-for" cash flows as optimistic cash flows, contrasted with expected cash flows. In essence, the promised principal and interest payments for bonds that are risky (i.e., have a non-negligible risk of default) are the bondholder's "hoped-for" cash flows. Thus, the term *promised* cash flow is synonymous with *hoped-for* cash flow.

The $Y_{Promised}$ is 10% in this instance. This promised yield to maturity is equal to the expected return to the investor only when the promised payment of $1,100 is made with certainty. In other words, $Y_{Promised}$ is the investor's expected rate of return (thus the firm's cost of debt capital) only when there is no risk of default.

When a firm's debt is subject to the risk of default, the promised yield to maturity *overstates* the expected yield to maturity to the debt holder. To see how the prospect of default drives a wedge between the promised and expected yields to maturity on debt, consider the simple case of a bond that matures in one year and pays the holder the $1,000 face value of the bond plus $100 in interest. The risky bond has a current price of $800 and promises to pay the holder $1,100 in principal and interest in one year. We calculate the promised yield to maturity by solving for the discount rate, $Y_{Promised}$, which satisfies the following discounted cash flow equation:

$$\begin{aligned}\begin{array}{c}\text{Bond Price}\\ \text{Today}\end{array} = \$800 &= \frac{\text{Promised Principal and Interest in One Year}}{(1 + Y_{Promised})^1}\\ &= \frac{\$1{,}100}{(1 + Y_{Promised})^1}\end{aligned}$$

Solving for the promised yield to maturity, we estimate it to be 37.5%.

Now assume that the likelihood of default by the issuer of the bond is 20% (i.e., there is an 80% probability that the bondholder will receive the full $1,100). Also, in the event of default, the debt holder will receive 60% of the promised $1,100 payment, or $660. The expected cash flow then is $(.80 \times \$1{,}100) + (.2 \times \$660) = \$1{,}012$. We can now calculate the expected yield to maturity using this expected cash flow as follows:

$$\begin{array}{c}\text{Bond Price}\\ \text{Today}\end{array} = \$800 = \frac{\text{Expected Bond Payment in One Year}}{(1 + Y_{Expected})^1} = \frac{\$1{,}012}{(1 + Y_{Expected})^1}$$

The estimate of $Y_{Expected}$ is only 26.5%. This represents the expected cost of the firm's debt capital, not the promised yield to maturity of 37.5% calculated above. The difference between the promised and the expected yield to maturity in this example is quite large, because the probability of default is quite high, and the recovery rate is 60%.

Default Rates, Recovery Rates, and Yields on Corporate Debt

In order to determine the expected return on a debt instrument, one needs estimates of both default rates and recovery rates. Moody's Investors Service provides historical research regarding both these rates and forecasts of future rates as well.

For example, Table A-1 summarizes the cumulative default rates on corporate bonds over the period 1983–2000. Note that there is a dramatic difference in default risk for investment- versus speculative-grade bonds. In fact, the cumulative default risk for investment-grade bonds is less than 2% even after 10 years, whereas the comparable rate for speculative-grade bonds exceeds 30%.

The important thing to note about Table A-1 is the following: For firms that issue investment-grade debt, the risk of default is so low that the difference between promised and expected yields is insignificant for practical purposes. However, for speculative-grade debt, issuers' default is much more likely, and it has a significant effect on the cost of debt capital.

What about bondholder recovery rates in the event of default? Estimates of expected recovery rates (like expected default rates) are based on historical experience,

Table A-1 Average Cumulative Default Rates for 1 to 10 Years—1983–2000

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Investment grade	0.05%	0.17%	0.35%	0.60%	0.84%	1.08%	1.28%	1.47%	1.62%	1.73%
Speculative grade	3.69%	8.39%	12.87%	16.80%	20.39%	23.61%	26.44%	29.04%	31.22%	32.89%

Source: Moody's Investors Service, *Default and Recovery Rates of Corporate Bond Issuers: 2000,* Global Credit Research (February 2001), Exhibit 42, p. 47.

and Moody's maintains records on recovery rates. Moody's approximates the recovery rate using the ratio of the market price to the par value of the bond one month after default. The average recovery rate in 2000 was 28.8% of par, a rate that was down from 39.7% a year earlier and the post-1970 average of 42%.[34]

Complicating the prediction process is the fact that recovery rates are inversely correlated with default rates. Very simply, higher default rates are often associated with lower recovery rates. However, for our purposes in the examples that follow, we assume a constant recovery rate.

Estimating the Expected Yield to Maturity for a Long-Term Bond

To illustrate how we might estimate the expected yield to maturity on a bond that faces the risk of default, consider the example posed in Table A-2. The bond has a 10-year maturity, a 14% coupon rate, and a $1,000 principal amount. This bond, which has a current price of $829.41, has a promised yield to maturity of 17.76%, which is of course substantially higher than the expected yield to maturity of 12.54%, which accounts for the probability of default and less than 100% recovery rate.[35]

To account for the probability of default, we list in the columns the cash flows that the bondholder receives in the event the bond defaults after each year up to 10 years. The probability of default for each of these default scenarios is found below each of the "Year" columns.[36] In addition, we assume that the recovery rate in the event of default is 50%, regardless of the year in which the default occurs. This means that in the event of default, the cash flow to the bondholders is $570 (i.e., the product of the $1,140 in promised interest plus principal multiplied by the estimated recovery rate of 50%).

Estimating the expected yield to maturity for the 10-year bond is a two-step process. The first step entails estimating the yield to maturity for the cash flows in each scenario. For example, if the bond were to default in the first year after its issuance, then the bondholder would receive a return one year later of only $570 (50% of the promised

[34] D. Hamilton, G. Gupton, and A. Berthault, *Default and Recovery Rates of Corporate Bond Issuers: 2000,* Moody's Investors Service Global Credit Research (February 2001), p. 3.

[35] The 17.76% promised YTM is consistent with the 12.75% spread over the YTM on 10-year Treasury bonds (5.02% YTM) for Caa/CCC bonds reported in Figure 3.

[36] The probability of default is equal to the incremental percent of below-investment-grade debt that defaults each year, as reported in Table A-1.

Table A-2 **Estimating the Expected Cost of Debt on a Long-Term Bond (Rated Caa/CCC)**

Coupon	14%	Bond rating Caa/CCC
Principal	$1,000.00	10-Year Treasury yield = 5.02%
Price	$ 829.41	
Maturity	10 years	
Recovery rate	50%	

		Default Cash Flows										Promised
	Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Cash Flow
	0	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)	(829.41)
	1	570.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
	2		570.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
	3			570.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
	4				570.00	140.00	140.00	140.00	140.00	140.00	140.00	140.00
	5					570.00	140.00	140.00	140.00	140.00	140.00	140.00
	6						570.00	140.00	140.00	140.00	140.00	140.00
	7							570.00	140.00	140.00	140.00	140.00
	8								570.00	140.00	140.00	140.00
	9									570.00	140.00	140.00
	10										570.00	1,140.00
Expected yield to maturity if default occurs in this year		−31.28%	−8.23%	0.98%	5.85%	8.80%	10.76%	12.12%	13.12%	13.87%	14.44%	**17.76%**
Probability of default in each year		3.69%	4.70%	4.48%	3.93%	3.59%	3.22%	2.83%	2.60%	2.18%	1.67%	67.11%
Weighted YTM = E(YTM) × Probability of default		−1.15%	−0.39%	0.04%	0.23%	0.32%	0.35%	0.34%	0.34%	0.30%	0.24%	11.92%
Average YTM based on expected cash flows		**12.54%**		**Expected YTM = Cost of debt**					**Promised YTM**			

principal plus interest), resulting in a rate of return or yield to maturity of −31.28% on his or her $829.41 investment.

Once the yield to maturity for each of the years in which default might occur (i.e., the default scenarios) is calculated, we can combine them to determine the expected yield to maturity for the bond. We consider a total of 11 different default scenarios corresponding to default in Years 1 through 10 plus the no-default scenario. The expected yield to maturity of the bond is equal to a weighted average of these possible yields, where the weights are equal to the probability of default in each year plus the probability of not defaulting at all. The resulting expected yield to maturity for this bond is 12.54%, which is substantially less than the promised yield to maturity of 17.76%.

In our example, we assumed a recovery rate of 50%. However, we reported earlier that the average recovery rates have been falling and are now approximately 30%. At this lower recovery rate, the expected YTM is only 9.13%. Consequently, lower recovery rates result in an even more dramatic difference between promised and expected YTM.

Estimating the Cost of Hybrid Sources of Financing—Convertible Bonds

Firms often issue what are referred to as **hybrid securities,** which have the attributes of debt and common equity or of preferred stock and common equity. For example, *convertible bonds and convertible preferred stock* represent a very important class of hybrid securities that give the holder the option to exchange, at his or her discretion, the bond or preferred stock for a prescribed number of common shares. Although these securities are not major sources of capital for investment-grade companies, they provide an important source of financing for many emerging high-tech companies and other smaller firms.

The value of a convertible security can be viewed as the value of the bond or preferred share plus the value of the conversion feature, which is technically a call option.[38] To illustrate, consider the case of a convertible bond whose value is defined by Equation A.1 below:

$$\begin{matrix}\text{Convertible}\\ \text{Bond Value}\end{matrix} = \begin{pmatrix}\text{Straight}\\ \text{Bond Value}\end{pmatrix} + \begin{pmatrix}\text{Call}\\ \text{Option Value}\end{pmatrix} \tag{A.1}$$

This "dual" source of value means that the cost of financing by issuing convertibles is a function of both the underlying security (bond or preferred stock) and the call option. Consequently, the cost of capital obtained by issuing convertible bonds can be thought of as a weighted average of the cost of issuing straight bonds and the cost of the exchange feature (call option), where the weights equal the relative contributions of the two components to the value of the security.

[38] We discuss options and their valuation in some detail beginning in section entitled "Validation in a Private Equity Setting". However, for now it is sufficient that we understand what a call option represents.

To illustrate how we might approach the estimation of the cost of convertible debt, consider the convertible debt issued by Computer Associates (CA) that is due March 15, 2013. The bonds sold for $1,066.19 on February 3, 2005, and paid a semiannual coupon payment based on an annual rate of interest equal to 5%. Each bond can be converted into 41.0846 shares of Computer Associates common stock beginning March 21, 2011. The convertible bond issue, like all of the firm's bonds, is rated BB+ by Standard and Poor's.

Before describing how to determine the appropriate cost of capital that should be ascribed to this convertible security, we first note that the promised yield to maturity on the bond, assuming that it is not converted, is only 1.61%. It should also be noted that the promised yield on the firm's straight debt is 4.25%. Clearly, the expected return on the convertible security exceeds the return on the straight debt, but by how much?

To estimate the cost of financing to Computer Associates from the convertible issue, we calculate a weighted average of the cost of straight debt and the cost of raising capital through the sale of the associated call option. We estimate the required rate of return on the convertible bonds using a three-step procedure:

Step 1: *Estimate the component values of the convertible bond issue: straight debt and the conversion call option.* Computer Associates' convertible bonds sold for $1,066.19 on the date of our analysis, so if we value the straight-debt component, we can calculate the value that the market is assigning to the call option component.

To value the straight-debt component, we need an estimate of the market's yield to maturity for straight debt that Computer Associates could issue today. The firm has a straight-debt issue outstanding with a similar maturity to the convertibles. The straight bond provides a market-based yield to maturity to the bondholders of 4.25%. Consequently, the straight-bond component of the convertible issue can be valued by discounting the promised interest and principal payments using 4.25%:

$$\begin{aligned}\begin{matrix}\text{Straight-}\\ \text{Bond Value}\end{matrix} &= \frac{\$25.00}{(1+.0425/2)^1} + \frac{\$25.00}{(1+.0425/2)^2} + \frac{\$25.00}{(1+.0425/2)^3} \\ &\quad + \frac{\$25.00 + \$1{,}000.00}{(1+.0425/2)^4} = \$1{,}014.24\end{aligned}$$

The value of the call option component, therefore, is $51.95 = $1,066.19 − 1,014.24.

Step 2: *Estimate the costs of the straight-debt and option components of the convertible bond issue.* We have already estimated the cost of the straight debt to Computer Associates using the yield to maturity on similarly rated bonds issued by the firm that have a similar term to maturity (i.e., 4.25%).[38] However, estimating the cost of the conversion feature (i.e., the conversion call option component) is more difficult and requires an understanding of call options. Because the call options can be

[38] Technically, the 4.25% yield to maturity on Computer Associates' straight debt is the *promised* yield; this rate overstates the expected cost of debt financing, as we discussed earlier.

viewed as a levered version of the firm's common equity, we know that the cost of capital associated with the option component of the convertible exceeds the cost of common equity of the issuing firm. In this example, we will assume that the cost of capital on the option component is 20%.

Step 3: *Calculate the cost of convertible debt.* Here we simply calculate a weighted average of debt and equity raised through the conversion feature. The weights attached to each source of financing reflect the relative importance (market values) of each:

	$ Value	**% of Value**
Straight bond	$1,014.24	95.1%
Conversion option	51.95	4.9%
Current bond price	$1,066.19	100.0%

Assuming that the conversion option has a required rate of return of 20% and using the 4.25% cost of straight debt, we estimate the cost of the convertible debt as follows:

	Fraction of Bond Value	**Cost of Capital**	**Product**
Straight bond	0.9513	4.25%	0.0404
Equity conversion option	0.0487	20.00%	0.0097
Cost of capital from convertible bonds =			$\boxed{5.01\%}$

Thus, we estimate that the cost of raising capital with convertible bonds is approximately 5%, which is higher than the cost of issuing debt but lower than the cost of issuing equity. This makes sense because convertibles are less risky than equity but more risky than straight debt.

Off-Balance-Sheet Financing and WACC

Following the high-profile financial scandals of the last decade, analysts have been made painfully aware of the importance of off-balance-sheet sources of financing—that is, sources of financing that are not listed among the firm's corporate liabilities but are nevertheless very real liabilities. Here we discuss special-purpose entities.

Special-Purpose Entities

The acronym SPE, which stands for *special-purpose entity,* has become synonymous with the wrongdoing that led to the collapse of Enron Corp. Unfortunately, Enron's use of SPEs has besmirched a very worthwhile concept that has been extremely valuable in a wide range of industries. Specifically, an SPE is an entity created by a sponsoring firm to carry out a specific purpose, activity, or series of transactions that are directly related to its specific purpose.[39]

[39] This discussion relies on Cheryl de Mesa Graziano, "Special Purpose Entities: Understanding the Guidelines," *Issues Alert, Financial Executives International* (January 2002).

The issue we want to discuss arises out of the fact that many SPEs are not consolidated with the firm's financial statements and, consequently, their assets and financial structures are off balance sheet. Should the capital structures of the off-balance-sheet SPEs be incorporated into the estimation of the firm's WACC? In Enron's case, many of the SPEs did not meet the rules prescribed for off-balance-sheet reporting, so the answer to the above question is obviously yes. However, where SPEs are truly separate entities (the technical term is *bankruptcy remote*), then their liabilities do not reach back to the founding firm and consequently are not relevant to the calculation of the firm's WACC.